

# Quantitative evaluation of empirical models of vortex-induced vibration of bridge decks through sectional model wind tunnel testing

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## ABSTRACT

The capabilities of some commonly used single-degree-of-freedom (SDOF) VIV models for bridge decks are examined through wind tunnel experiment. The VIV responses of a sectional deck model at different Scruton numbers are measured. An instantaneous identification method is adopted for obtaining the properties of aeroelastic damping and stiffness of the VIV system. The simulated aeroelastic damping by these models as well as the predicted VIV responses of the deck at different Scruton numbers are compared with experiments. The results indicate that the more accurate the simulated aeroelastic damping by an empirical model, the smaller the prediction errors by this model will be. Through directly fitting the aeroelastic damping, the prediction errors by an empirical model can be further reduced.

## 1. Introduction

Vortex-induced vibration (VIV) is a large amplitude oscillation phenomenon, which often occurs at low and moderate wind velocities [1–4]. VIV can influence the usability of the structure or even cause fatigue failures to some important structural elements. Table 1 summarizes the bridges that have experienced VIV. As can be seen, almost all bridge types are sensitive to VIV. With the ever-growth of span length, bridges are becoming more flexible and thereby more susceptible to VIV. Careful attentions should be paid to assess the VIV of the bridge in the design stage.

To assess the VIV of a bridge, the most direct way is to conduct wind tunnel tests on a full three-dimensional (3D) aeroelastic model that has similar dynamics and configurations with this bridge. However, the scale ratio of a full aeroelastic model in wind tunnel is usually quite small. The Reynolds number effect and scale effect (such as the manufacture errors of crash barriers, railings, and/or maintenance rails) significantly influence the accuracy of the experimental results. For example, Li et al. [1] compared the VIV responses of the Xihoumen Bridge between field measurement and sectional model test in wind tunnel, and found that the response of sectional model is much smaller than that of field measurement. The reasons were attributed to the Reynolds number effects [5,6]. Frandsen [7] compared the field measurement of Great Belt East Bridge to those of full aeroelastic model (1:200) and sectional models (1:60 and 1:80). It indicates that both the

VIV velocities and amplitudes of the 1:200 full aeroelastic model are much lower than those of field measurement and sectional model test. Thus, in real practice, wind tunnel test on VIV is normally conducted on sectional models or even large-scale sectional models, which have similar modal dynamics (i.e. modal mass, modal frequency and modal damping) with a certain mode of the bridge.

The VIV assessment for a bridge then consists of two steps: firstly, the fluid-structure interaction during VIV is modeled by using empirical models; secondly, the empirical models are combined with structural finite element model to take the influence of modal shapes and spatial coherence of VIV forces into account [8–11].

From the above, a model that can reasonably simulate the fluid-structure interactions during VIV is important for predicting the VIV of a bridge. However, the mechanism behind VIV of bluff bodies is quite complex and the VIV system may show abnormal dynamic properties under certain circumstances, such as hysteresis of amplitude branches, phase jump phenomenon and bifurcation [12–14]. A rigorous mathematical modeling of VIV needs simultaneously solving the Navier-Stokes (N-S) equation with three dimensional (3D) grids and arbitrarily shaped moving boundaries, which will be difficult to implement in the foreseeable future. Existing models mainly focus on reproducing the major dynamic properties of the VIV system, such as the limit cycle oscillation (LCO).

Following the concept of Bishop and Hassan [15], the modeling of bridge deck VIV was first conducted by Scanlan [16]. The bridge and

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**Table 1**  
Prototype bridges that experienced VIV.

| Bridge Type         | Location | Bridge Name                               | Main Span (m) | Wind Velocities (m/s) | Maximum Amplitude(cm) |
|---------------------|----------|-------------------------------------------|---------------|-----------------------|-----------------------|
| Cable-stayed bridge | England  | Kessock Bridge [30]                       | 240           | 23–25                 | 11                    |
|                     | England  | Second Severn Bridge [31]                 | 456           | 15–20                 | 27                    |
|                     |          |                                           |               |                       | (RMS: 19.2)           |
|                     | Japan    | Ishikari Kako Bridge [32]                 | 160           | 10                    | 10                    |
|                     | Korea    | Jindo Bridge [33]                         | 340           | 9.8–11.5              | 10                    |
| Suspension bridge   | Canada   | Wye Bridge [34]                           | 235           | 7–8                   | 3.6                   |
|                     | America  | Fred Hartman Bridge [35]                  | 380           | 8–12                  | –                     |
|                     | China    | Xihoumen Bridge [1,2]                     | 1650          | 5–12                  | 24                    |
|                     | Denmark  | Great Belt East Bridge [36]               | 1624          | 5–12                  | 32                    |
|                     | America  | Old Tacoma Narrows Bridge [37]            | 853           | < 18.4                | –                     |
| Girder bridge       | America  | Deer Isle Bridge [38]                     | 329           | 9.11                  | 5.5                   |
|                     | Brazil   | Rio-Niteroi Bridge [39]                   | 300           | 15–17                 | 25                    |
|                     | Japan    | Trans-Tokyo Bay Bridge [3]                | 240           | 16–17                 | 54                    |
|                     | Russia   | Volga Bridge [40]                         | 155           | 11.6–15.6             | 65–70                 |
|                     | Denmark  | Great Belt East Bridge Approach Span [41] | 193           | 16–20                 | 10                    |

surrounding flow field were treated as a unit system with nonlinear damping properties, i.e. the single-degree-of-freedom (SDOF) empirical model. Since then, several attempts have been conducted towards this direction to more reasonably reflect the energy transfer from the flow field to the deck during VIV [8,17–22]. This kind of models are easy to implement, since the model parameters are identified through measured deck responses in wind tunnel. With the development of experimental facilities, direct measurement of VIV force in wind tunnel becomes available through forced or free vibrations. The modeling of VIV force was then proposed by using equivalent nonlinear oscillators [23], polynomial functions [24], or Volterra Series [25,26]. The accuracy of these models depends on the precision of VIV force data and elaborate experimental facilities (for example, high-precision force balance), which may not always be available in wind tunnels. Quite recently, data-driven approach through machine learning (ML) scheme was also proposed to model the VIV of bridge [27,28]. This method can directly build the relationship between wind properties (such as wind speed, direction and spatial coherence) and bridge responses. But the efficiency of this approach relies on the scale and information richness of the database.

Since SDOF empirical models are easy to implement, they are still widely adopted. However, the application of SDOF empirical models is challenged by two difficulties. The first challenge is the influence of Scruton number ( $S_c = 2m\xi/(\rho D^2)$ ,  $m$  is the mass of the structure per unit length,  $\rho$  is air density,  $D$  is deck height and  $\xi$  is structural damping ratio). The identified model parameters of empirical models vary with Scruton number, which indicates that once the Scruton number of the prototype bridge is different from that on which the model parameters were identified, the feasibility of the empirical model fails.

The second challenge is that the calculated “mode shape correction factor” (i.e. the ratio of VIV amplitude of the prototype bridge to that of sectional model) varies with different empirical models, even though all the other conditions keep the same. For example, if perfectly correlated vortex excitation force and sinusoidal modal shape are assumed, the “mode shape correction factor” is  $4/\pi$  by Scanlan’s linear semi-empirical model, while, it is  $2/\sqrt{3}$  by Scanlan’s nonlinear semi-empirical model [29]. This indicates that even the model parameters are identified on the exact Scruton number of the prototype bridge, the predicted prototype response of the bridge may not necessarily be trustworthy, since the result depends on the property of the model adopted for calculation.

From the above, the property of a semi-empirical model is important for predicting the prototype response of a bridge. Since SDOF empirical models treat the bridge and surrounding flow field as a unit nonlinear system, the accuracy of the model in reproducing the nonlinear property (especially the aeroelastic damping and stiffness) of the VIV system can be adopted as a criterion for evaluating the capability of this model.

In this study, the capability of some commonly used SDOF models is evaluated based on wind tunnel experiment. An instantaneous identification method is adopted for obtaining the properties of aeroelastic damping and stiffness of the VIV system. The simulated aeroelastic damping as well as the predicted VIV responses of the deck at different Scruton numbers are compared with experimental results. It is noticeable that the goal of this research is not to validate the reasonability of the “negative damping” assumption behind the SDOF empirical models, but to quantitatively evaluate the capabilities of commonly used empirical models in reproducing the nonlinear property of the VIV system, the prediction errors by these models once the Scruton number varies, and to propose some practical guidelines for further improving the accuracy of the empirical models.

## 2. Nonlinear properties of VIV system

### 2.1. Signal processing algorithms

The major property of the VIV system is self-excited and self-limited oscillation. This indicates that the dynamics of the system vary with its vibration amplitude. For such a nonlinear system, the properties of frequency and damping versus amplitude are important for assessing its nonlinear characteristics. These properties can be identified through decay- or grow-to-resonance tests in wind tunnel. Since the damping, stiffness and amplitude keep changing during the test process, their instantaneous values should be identified firstly.

The instantaneous frequency of the VIV system is identified by using the Direct Quadrature (DQ) method proposed by Huang et al. [42]. This method consists of two steps: first, Empirical mode decomposition (EMD) technique is adopted to decompose any given signal as a set of nearly monocomponent signals (defined as Intrinsic Mode Function, IMF); then the DQ method is applied to the normalized IMFs to get an exact instantaneous frequency. This approach can totally eschew the Hilbert transform (HT), which only produces meaningful instantaneous frequency if the conditions of Bedrosian theorems are met.

Since VIV is a SDOF phenomenon, the measured decay- or grow-to-resonance displacement is narrow banded. The second step is thus directly applied to the measured data. The decay- or grow-to-resonance displacement can be expressed as:

$$y(t) = A(t)\cos[\varphi(t)] \quad (1)$$

where  $A(t)$  is time-varying amplitude and  $\varphi(t)$  is time-varying phase of motion.

The instantaneous frequency of the SDOF system is defined as the time derivative of  $\varphi(t)$ , which can be obtained through the following procedures:

- 1) taking the absolute values of  $y(t)$  and extracting all the local maximum points of  $|y(t)|$ ;
- 2) connecting all these points through a cubic spline curve  $e_1(t)$  and normalizing  $y(t)$  with  $e_1(t)$ :

$$N_1(t) = \frac{y(t)}{e_1(t)} \quad (2)$$

- 3) repeating the above normalizing approach until all the values of  $N_n(t)$  are less than or equal to unity:

$$\begin{aligned} N_2(t) &= \frac{N_1(t)}{e_2(t)} \\ &\vdots \\ N_n(t) &= \frac{N_{n-1}(t)}{e_n(t)} \end{aligned} \quad (3)$$

- 4) the VIV displacement of the deck is then decomposed into the envelope part  $A(t)$  and the carrier part  $\cos[\varphi(t)]$ :

$$\begin{aligned} A(t) &= e_1(t)e_2(t)\cdots e_n(t) \\ \cos[\varphi(t)] &= N_n(t) \end{aligned} \quad (4)$$

- 5) the phase of motion  $\varphi(t)$  is then obtained:

$$\varphi(t) = \arctan \frac{\sqrt{1 - N_n^2(t)}}{N_n(t)} \quad (5)$$

- 6) taking time derivative of  $\varphi(t)$ , the instantaneous frequency is obtained:

$$\omega(t) = \frac{d\varphi(t)}{dt} \quad (6)$$

VIV system is a weak nonlinear system, for such a system, as illustrated by Fig. 1, the variation of vibration amplitude and frequency in a single period of time is quite small as compared with their magnitudes. Thus, the values of amplitude and frequency for each single period can be treated as constants, which equal to the mean values of the identified instantaneous envelopes and frequencies within that period, respectively.

As to the damping ratio, it can also be treated as a constant within each single period. For the  $j$ th period of vibration, the envelope of displacement can be expressed as:

$$A(t) = A_{j-1} \exp(-\xi_j \omega_j t) \quad (7)$$

where  $A_{j-1}$  is the initial point of the envelope for the  $j$ th period of vibration;  $\xi_j$  is damping ratio of the  $j$ th period;  $\omega_j$  is natural frequency of the  $j$ th period, which can be obtained by:

$$\omega_D = \omega_j \sqrt{1 - \xi_j^2} \quad (8)$$

in which,  $\omega_D$  is actual vibration frequency of the  $j$ th period.

Combing Eq. (8) with Eq. (7), we have:

$$\ln \frac{A_{j-1}}{A_j} = \frac{2\pi\xi_j}{\sqrt{1 - \xi_j^2}} \quad (9)$$

where  $A_j$  is the last point of the envelope for the  $j$ th period of vibration. The damping ratio  $\xi_j$  can then be calculated by:

$$\xi_j = \frac{1}{\sqrt{4\pi^2 + \ln(A_{j-1}/A_j)}} \ln \left[ \frac{A_{j-1}}{A_j} \right] \quad (10)$$

After obtaining the amplitude, frequency and damping ratio of each period, the properties of the frequency and damping of the VIV system versus structural amplitude can be obtained.

Experimental results indicate that frequency components in the VIV force are integral multiples of deck's vibration frequency [23,24]. To simulate the measured VIV forces, equivalent nonlinear oscillators [23] and modified SDOF empirical models [24] were proposed, which are essential polynomial functions of structural displacement and/or velocity. In this sense, the properties of frequency and damping of the VIV system versus amplitude can also be modeled by using polynomial functions:

$$\xi(A) = \sum_{i=1}^{N+1} b_i (A)^{i-1} \quad (11)$$

$$\omega(A) = \sum_{i=1}^{M+1} c_i (A)^{i-1} \quad (12)$$

where  $N$  and  $M$  denote the highest-order of polynomials for damping and frequency, respectively;  $A$  is deck amplitude;  $b_i$  and  $c_i$  are coefficients to be determined through fitting of experimental results.

## 2.2. Additional aeroelastic effects on the deck

Ideally, the suspension system in wind tunnel is linear, the frequency and damping of which are constants. The additional aeroelastic stiffness and damping on the deck during VIV can be calculated through deducting the constant values of the suspension system from the total VIV system. However, experimental results indicate that the frequency and damping of the suspension system in wind tunnel may also vary versus structural amplitude [43,44]. This nonlinearity may be due to the interaction between sectional model and its surrounding air in wind tunnel, the friction between joints and interfaces of the suspension arms, and/or the internal friction within the suspension springs [43].

To exclude the influence of suspension system and to extract the additional aeroelastic effects on the deck during VIV, the frequency and damping of the suspension system in still air can also be modeled by polynomial functions of vibration amplitude, then the additional aeroelastic damping and stiffness are calculated by

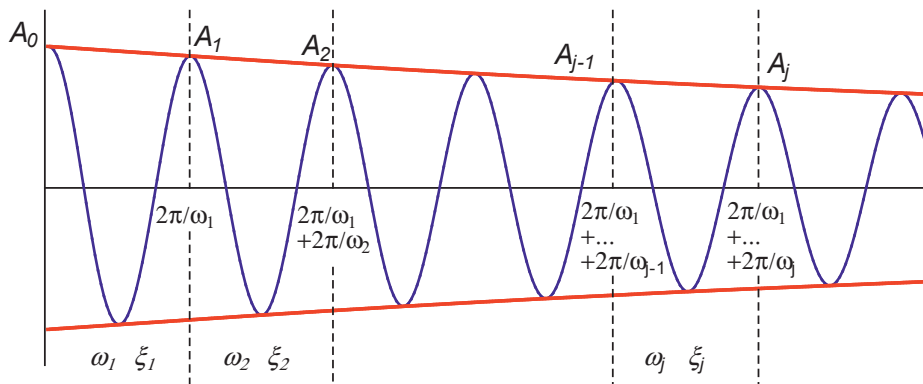


Fig. 1. Diagram of the vibration for a SDOF nonlinear system.

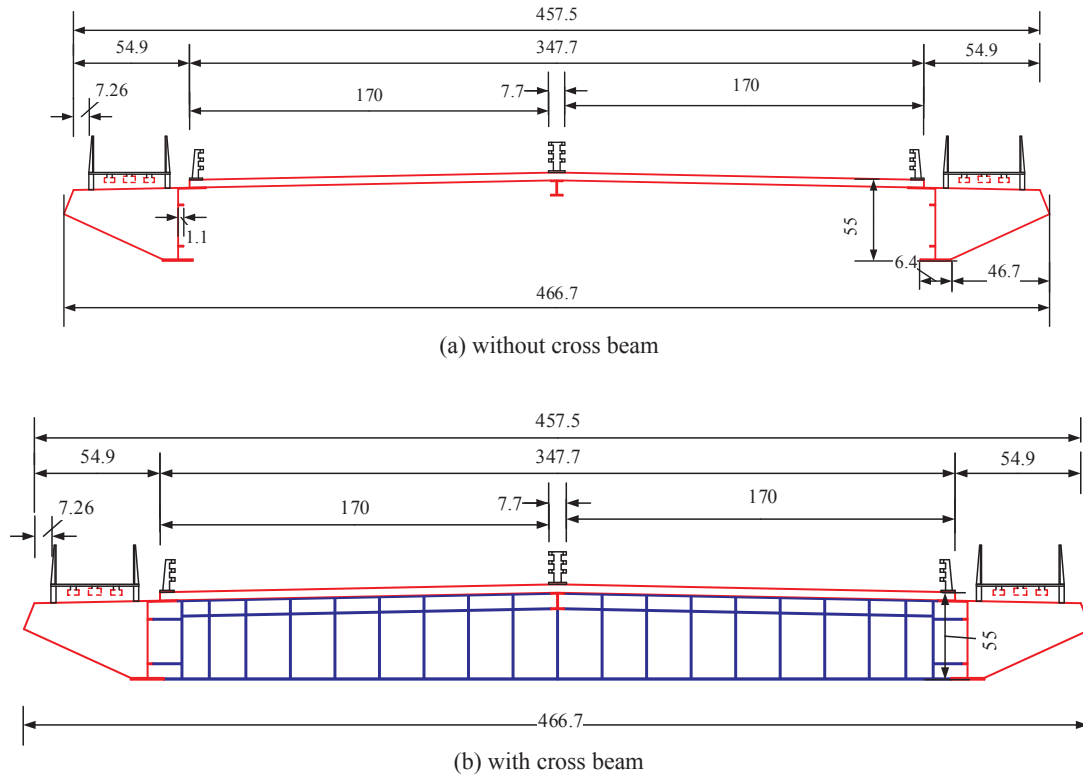


Fig. 2. Cross section of the deck model in wind tunnel (unit: mm).

$$\xi_{air}(A) = \xi(A) - \xi_{still}(A) \quad (13)$$

$$K_{air}(A) = m\omega^2(A) - m\omega_{still}^2(A) \quad (14)$$

where  $\xi_{still}$  and  $\omega_{still}$  are damping and frequency of the suspension system in still air, respectively;  $m$  is the mass of the sectional model per unit length.

### 3. Experimental arrangements

#### 3.1. Simulation of different Scruton numbers

Sectional model test was conducted in TJ-2 wind tunnel in Tongji University. The deck model is an open-section deck with two wind fairings on each side (the leading edge and the trailing edge), as shown in Fig. 2. This kind of deck shape is widely adopted for cable-supported bridges with steel-concrete composite beams. Detailed structural properties of the sectional model are summarized in Table 2.

To simulate different Scruton numbers, an external damping device was adopted. This device consists of two parts: an oil box and an aluminum stick. One end of the stick was fixed to the arm of the suspension system and the other end was connected to a plate that submerged in the oil box. By adjusting the width or length of the submerged plate, different values of external damping can be obtained. The schematic diagram of the experimental arrangement and the external damping device are illustrated by Fig. 3(a) and Fig. 3(b), respectively. Three different plate lengths were adopted during the experiment. Thus, along with the original suspension system (without external damping device), four damping conditions were simulated.

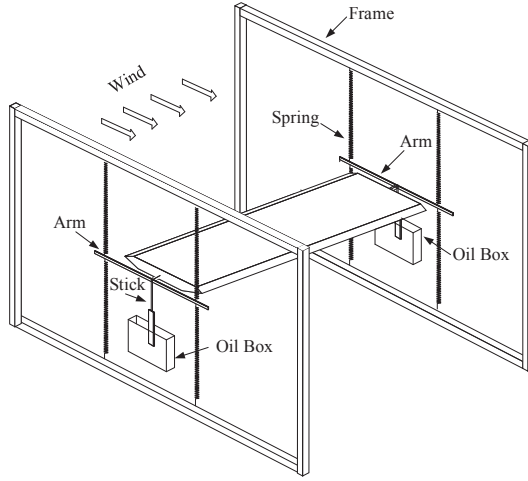
Table 2  
Structural properties of the sectional model.

| Height $D$ (mm) | Width $B$ (mm) | Length $L$ (mm) | Mass $m$ (kg/m) | Mass moment of inertia $I_m$ (kg·m <sup>2</sup> /m) | Vertical frequency $f_v$ (Hz) | Torsional frequency $f_t$ (Hz) |
|-----------------|----------------|-----------------|-----------------|-----------------------------------------------------|-------------------------------|--------------------------------|
| 55              | 466.7          | 1740            | 11.361          | 0.448                                               | 4.185                         | 8.95                           |

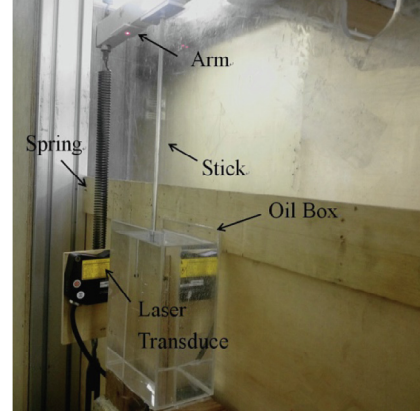
Figs. 4 and 5 show the identified instantaneous damping and frequency of the suspension system in still air. In this figure, Condition (1) denotes the original suspension system, Conditions 2–4 denote the suspension systems with different external damping. For each damping condition, two samples were measured to validate the repeatability of the results. As can be seen, the damping of the suspension system in wind tunnel varies versus structural amplitude ( $A/D$  is non-dimensional amplitude with respect to deck height). This indicates that the suspension system in still air has a nonlinear characteristic. Its influence should be deducted from the total damping of the VIV system to obtain the additional aeroelastic damping on the deck. As to the frequency of the suspension system, it slightly decreases versus structural amplitude. However, this trend is quite small. For simplification, the frequency of the suspension system is assumed to a constant (takes the mean value of the identified results).

Fig. 6 shows the steady-state VIV amplitudes of the sectional model at different damping conditions. The Scruton number in Fig. 6 is defined as  $S_c = 2m\bar{\xi}_{still}/(\rho D^2)$ , where  $\bar{\xi}_{still}$  takes the mean value of identified instantaneous damping in Fig. 4. As can be seen, the amplitude of the deck decreases with the increase of Scruton number. For  $U/f_n D = 20.92$ , the VIV phenomenon disappears at Condition (3) and Condition (4). This is because the increase of Scruton number not only influences deck responses but also the range of lock-in region. The influence of Scruton number on VIV properties of structures has been widely investigated, and will not be further discussed in this study.

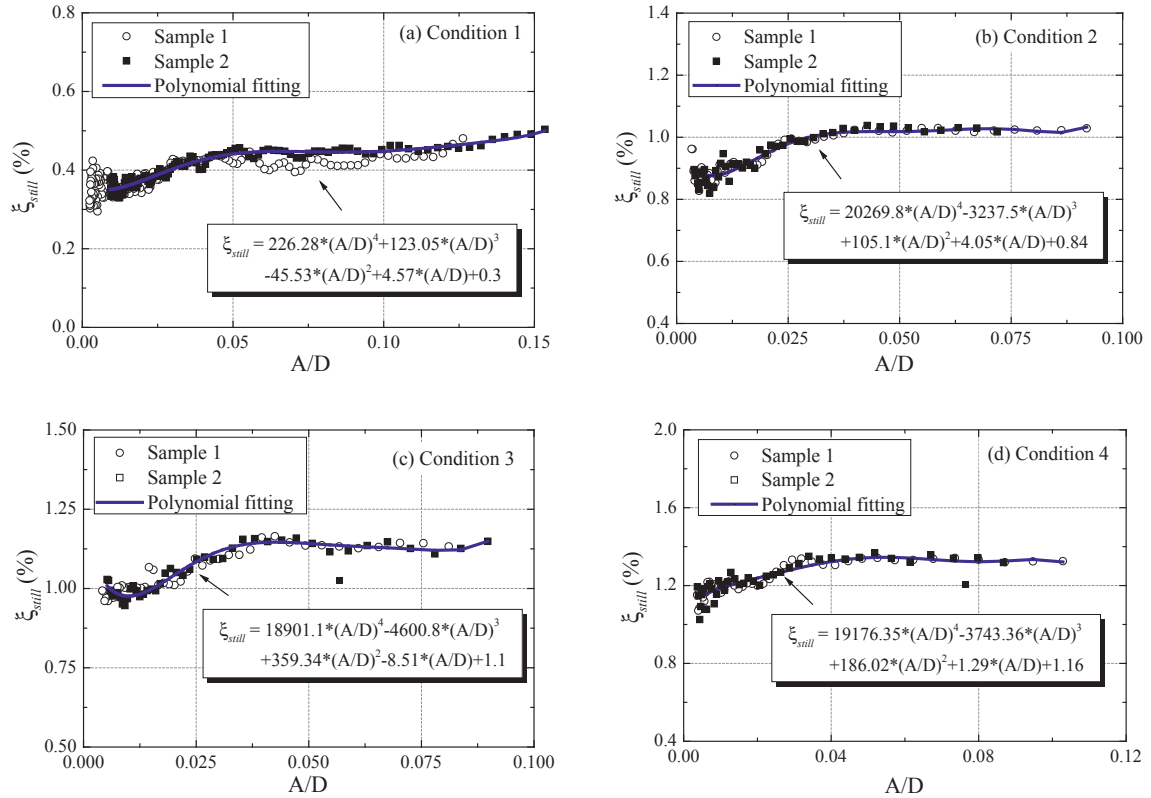
Fig. 7 shows the identified damping and frequency of the VIV system on damping Condition (1). The results are obtained by using both decay- and grow-to-resonance displacements. As can be seen, the



(a) experimental arrangement



(b) external damping device

**Fig. 3.** Schematic diagram of experimental arrangement and the external damping device.**Fig. 4.** Instantaneous damping of the suspension system at different damping conditions (cycles and squares denote two individual samples on each damping condition; solid lines are polynomial fittings of the samples).

damping curves show a nonlinear characteristic versus structural amplitude. Besides, the curve shapes vary with wind velocity. This indicates that the damping of the VIV system has both nonlinear and unsteady features. Similar with the results in still air, the variation of frequency versus structural amplitude is quite small. Thus, the frequency of VIV system is also assumed to be a constant.

### 3.2. Evaluation of aeroelastic effects at different Scruton numbers

The additional aeroelastic effects can be obtained through deducting the properties of suspension system in still air from those of the total VIV system. To validate the accuracy of this approach, the

aeroelastic damping identified on different damping conditions are compared.

Fig. 8 shows the total damping of the VIV system on Condition (1) and Condition (2). The solid and dashed lines are fourth-order polynomial fittings of these experimental results. As can be seen, the shapes of damping curves on these two conditions are quite similar with each other.

Fig. 9 compares the identified additional aeroelastic damping on these two conditions. The additional aeroelastic damping are obtained through deducting the properties of the suspension system in still air (modeled by polynomial functions in Fig. 4) from those of the total VIV system (modeled by polynomial functions in Fig. 8). As can be seen, the



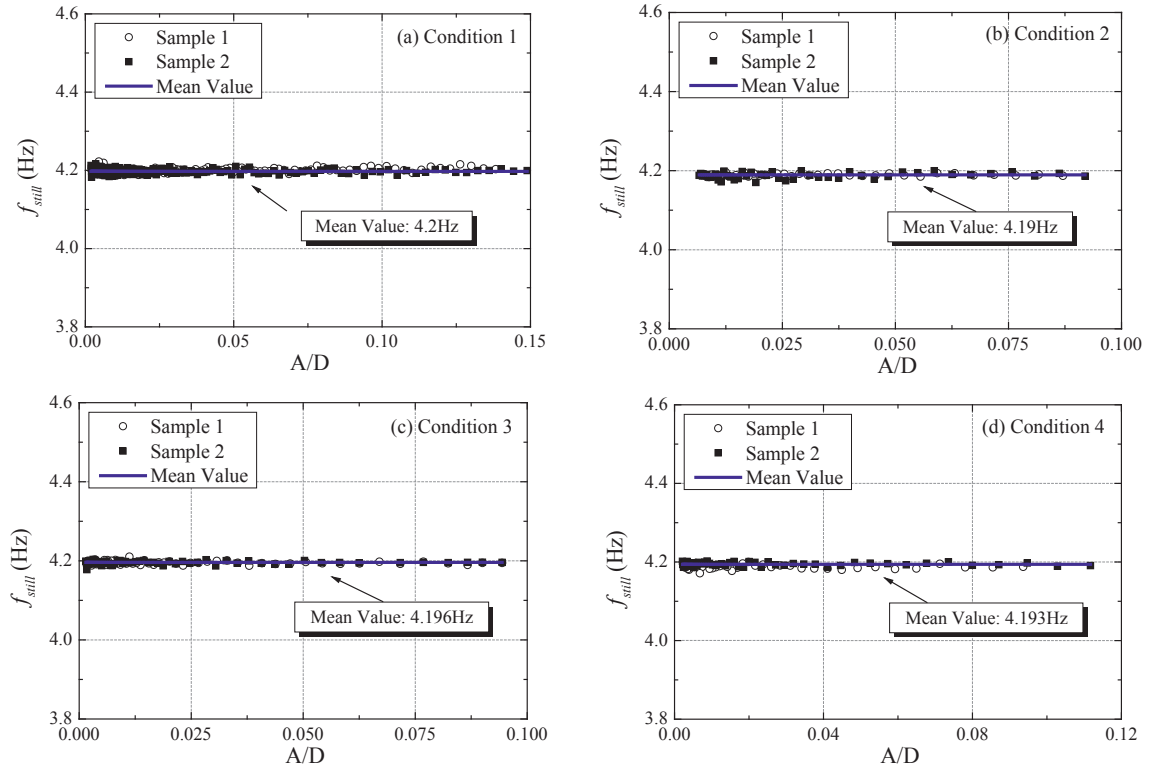


Fig. 5. Instantaneous frequency of the suspension system at different damping conditions (cycles and squares denote two individual samples on each damping condition; solid lines are mean values of the samples).

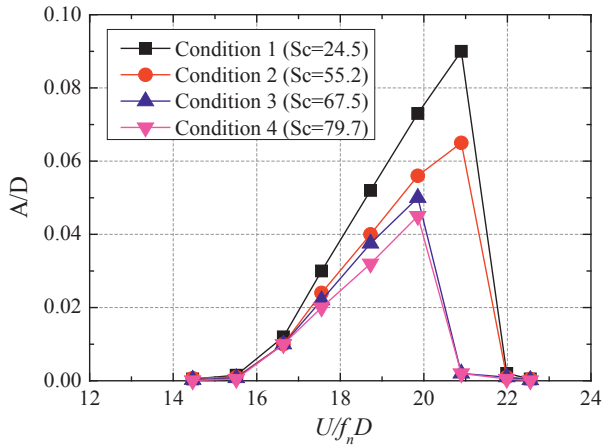


Fig. 6. VIV responses of the deck at different damping conditions.

additional aeroelastic damping on these two conditions are almost identical. This indicates that the proposed method can well extract the additional aeroelastic effects of the VIV system through wind tunnel experiment. The identified results can be adopted as a criterion for assessing the capability of some commonly used VIV models in the following section.

#### 4. Quantitative evaluation of empirical models

In this section, the capabilities of some commonly used semi-empirical models are evaluated. Firstly, the model parameters are identified through experimental data. Then, the simulated aeroelastic damping by this model is compared with the experimental result to evaluate its capability in reproducing the nonlinear property of the VIV system. Finally, the model is adopted to predict VIV responses of the

deck at different Scruton numbers to quantify the prediction errors of this model.

It is noticeable that there may exist several different methods for model parameter identification. For example, the parameters in the Scanlan's nonlinear empirical model can be identified through fitting the envelop of displacement [8] or through nonlinear minimization of numerical solution of the governing equation and experimental result [20]. However, the slight difference between parameter identification methods will not influence the nonlinear property of the model. Thus, only the most commonly used method is adopted for parameter identification.

##### 4.1. Periodic equivalence of semi-empirical models

It should be noted that the VIV force in empirical models is a function of structural acceleration, velocity and/or displacement, while the experimentally identified aeroelastic damping and stiffness is a function of structural amplitude. To compare with experimental result, the VIV force models should be transformed to the function of structural amplitude. For a SDOF VIV model, the governing equation can be generally written as [8]:

$$m(\ddot{y} + 2\xi_n\omega_n\dot{y} + \omega_n^2y) = F(\ddot{y}, \dot{y}, y, U, t) \quad (15)$$

where  $m$  is deck mass per unit length;  $\xi_n$  and  $\omega_n$  are mechanical damping and frequency of the structure, respectively;  $y$  is vertical motion of the deck;  $F$  is a general fluid-induced forcing function per unit length, i.e. the VIV force.

The transformation of  $F$  is based on the concept of periodical equivalence of energy. The total net work done by the VIV force and structural mechanical damping in each single period (i.e. the net energy transferred from the flow field to the deck) can be replaced by an equivalent one done by the total damping of the VIV system:

$$\int_0^{2\pi/\omega} 2\xi_{eq}\omega_n\dot{y}\cdot\dot{y}dt = \int_0^{2\pi/\omega} [2\xi_n\omega_n\dot{y} - F(\ddot{y}, \dot{y}, y, U, t)]\cdot\dot{y}dt \quad (16)$$

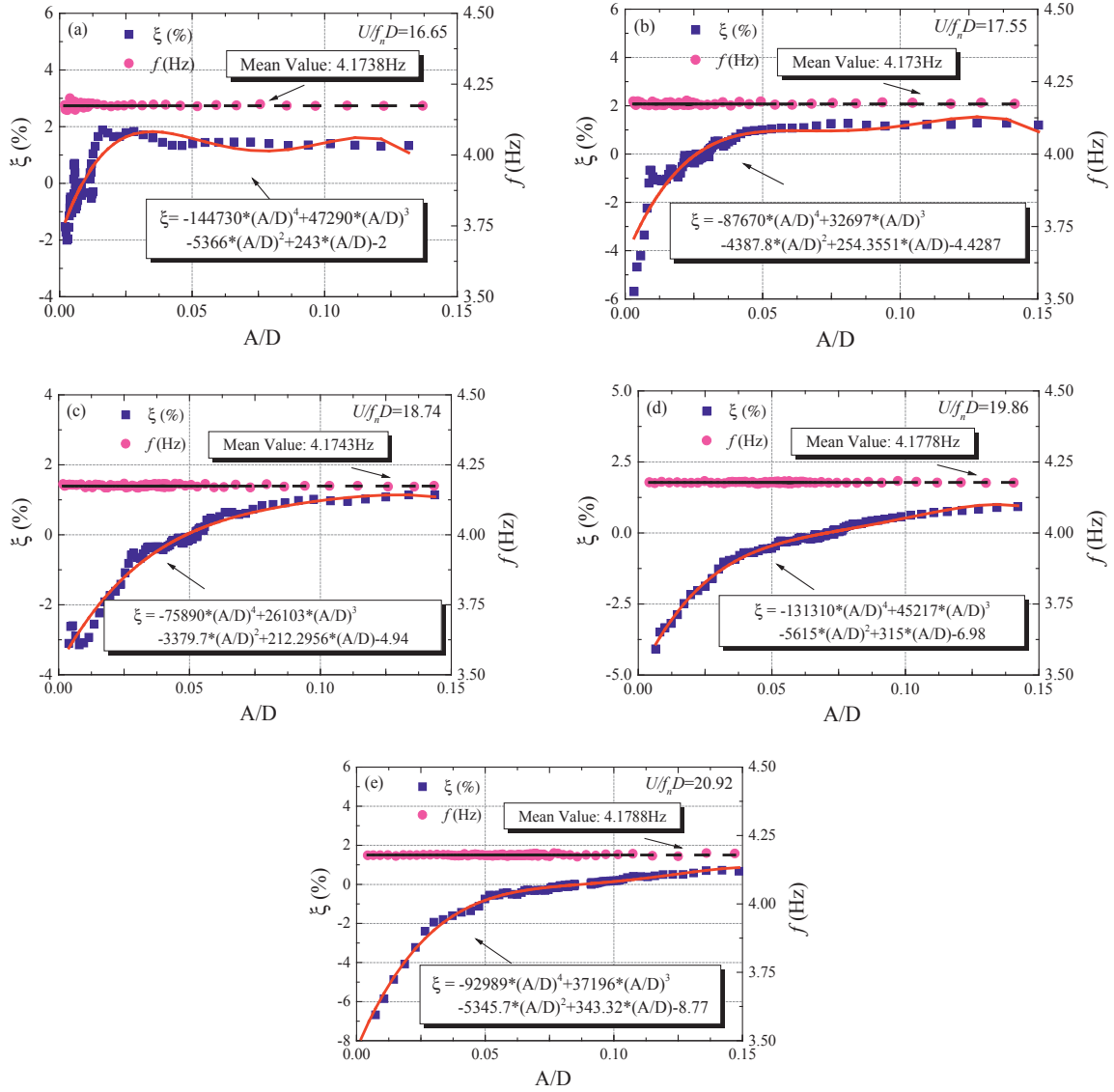


Fig. 7. Instantaneous frequency and damping of the total VIV system on Condition (1) (squares denote the damping; cycles denote the frequency; solid lines are polynomial fittings of damping; dashed lines are mean values of frequency).

where  $\omega$  is vibration frequency of the VIV system;  $\xi_{eq}$  is the periodical equivalent damping of the total VIV system.

The displacement and velocity of the VIV system in a single period of vibration can be expressed as:

$$y(t) = A \cos(\omega t); \dot{y}(t) = -\omega A \sin(\omega t) \quad (17)$$

where  $A$  is vibration amplitude in this period.

Substituting the above equations into Eq. (16), the equivalent damping of a VIV force model in a single period of vibration (as a function of structural amplitude in that period) can be obtained. After obtaining the periodic equivalent damping of the whole decay- and grow-to-resonance process, the properties of equivalent damping of an empirical model versus structural amplitude can be obtained. These results can then be compared with the experimentally identified ones.

#### 4.2. Linear empirical model by Scanlan

The linear semi-empirical model has the following form [17]:

$$m(\ddot{y} + 2\xi_n \omega_n \dot{y} + \omega_n^2 y) = \frac{1}{2} \rho U^2 D \left( KH_1^* \frac{\dot{y}}{U} - C_L \sin \omega_s t \right) \quad (18)$$

where  $\rho$  is air density;  $D$  is deck height;  $U$  is wind velocity;  $\omega_s$  is vortex-shedding frequency which is practically identical with the structural frequency  $\omega$  in the lock-in region;  $K = \omega D/U$  is reduced vibration frequency;  $C_L$  and  $H_1^*$  are magnitude of the forcing term and aeroelastic damping term, respectively, which should be determined through experiment. The other symbols are identical with those defined above.

By introducing non-dimensional time  $s = Ut/D$  and non-dimensional displacement  $Y = y/D$ , the above equation can be transformed to

$$Y'' + 2\xi_n K_n Y' + K_n^2 Y = \frac{\rho D^2}{2m} [KH_1^* Y' - C_L \sin Ks] \quad (19)$$

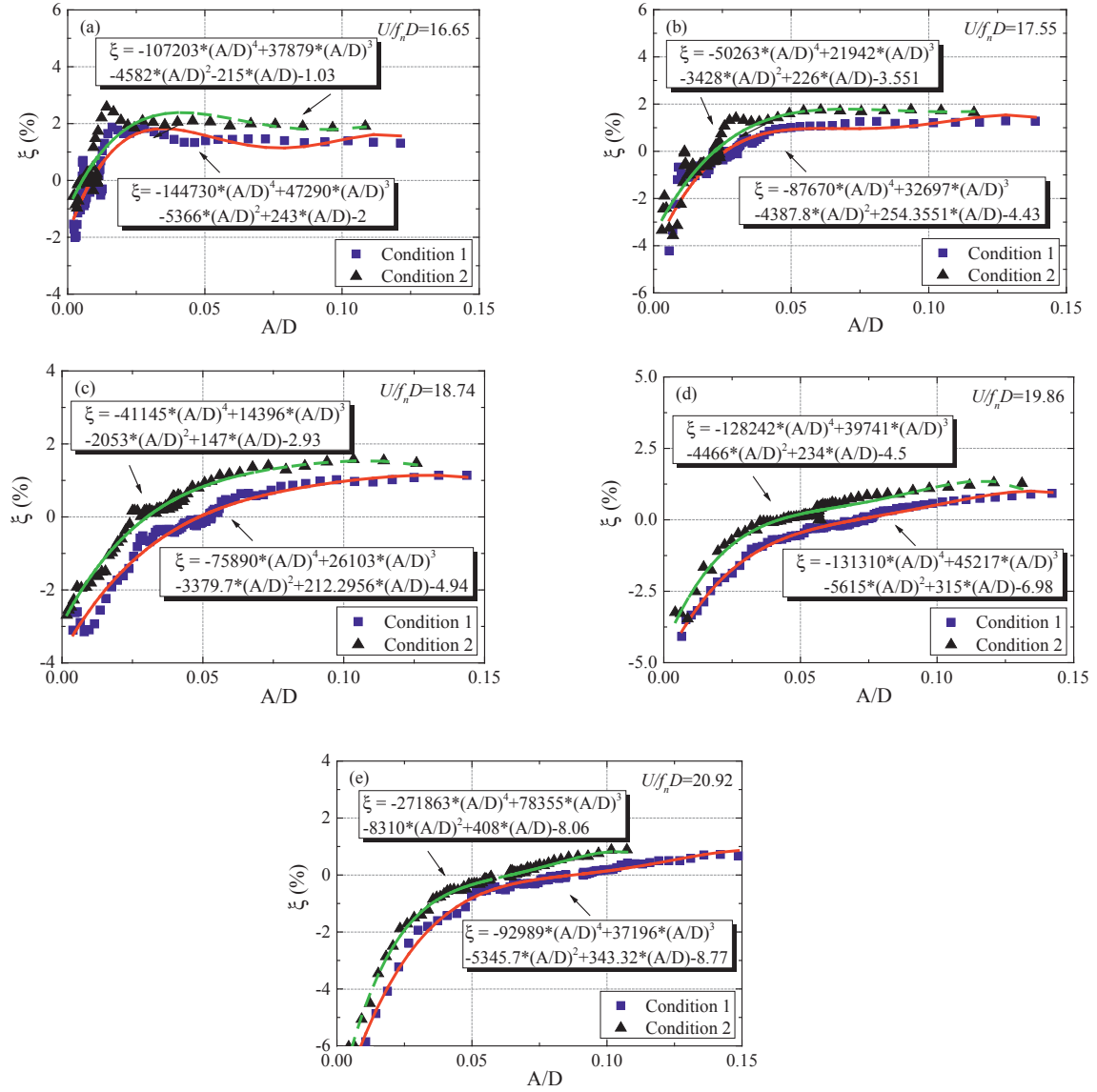
where  $K_n = \omega_n D/U$  is reduced structural natural frequency.

According to Scanlan [17], the model parameters  $H_1^*$  and  $C_L$  can be identified through decay-to-resonance displacement:

$$H_1^* = \frac{4m}{\rho D^2} (\xi_n - \xi); C_L = \frac{4m\xi K^2 A_s}{\rho D^2} \quad (20)$$

where  $A_s$  is steady-state VIV amplitude;  $\xi$  is total damping of the VIV system, which can be calculated by

$$\xi = \frac{1}{2n\pi} \ln \frac{A_0}{A_n} \quad (21)$$



**Fig. 8.** Total damping of the VIV system on Condition (1) and Condition (2) (squares and triangles are experimental results; solid lines are polynomial fittings of these experimental results).

in which,  $A_0$  and  $A_n$  are initial and  $n$ th cycle's amplitudes of the deck, respectively. The total damping of the VIV system can be identified at  $Ks = 2\pi$  ( $n = 1$ ),  $10\pi$  ( $n = 5$ ) and  $20\pi$  ( $n = 10$ ), respectively, and then take the average value of the identified results [17].

Taking the VIV response of the deck at  $U/f_n D = 19.86$  on Condition (1) as an illustration. The model parameters are obtained, as summarized in Table 3. In this process, the mechanical damping of the suspension system  $\xi_n$  adopts the averaged value of  $\xi_{still}$  in Fig. 4.

Fig. 10 compares the simulated decay-to-resonance displacement by the model and the experimental result. As can be seen, the simulated steady-state VIV amplitude compares well with the experimental result. However, the transient part of displacement varies a lot with the experimental result.

To investigate the difference between simulated and experimental results, the periodic equivalent damping of the linear empirical model is obtained by using the procedure in Section 4.1. For a single period of vibration, the displacement and velocity of the deck can be expressed as:

$$Y(s) = \frac{A}{D} \cos(Ks) \text{ and } Y'(s) = -K \frac{A}{D} \sin(Ks) \quad (22)$$

Substituting the above equation into Eq. (16), the periodic equivalent damping of the linear empirical model is obtained:

$$\begin{aligned} \xi_{eq} &= \xi_n - \frac{m_r H_1^*}{4} - \frac{C_L m_r}{4K_n^2} \left( \frac{A}{D} \right)^{-1} \\ \xi_{air} &= \xi_{eq} - \xi_n = -\frac{m_r H_1^*}{4} - \frac{C_L m_r}{4K_n^2} \left( \frac{A}{D} \right)^{-1} \end{aligned} \quad (23)$$

where  $m_r = \rho D^2/m$  is mass ratio.

Fig. 11 compares the periodic equivalent damping of the linear empirical model with the experimental result. It is noticeable that the mechanical damping of the suspension system is not constant, to deduct its influence, the aeroelastic damping is calculated by  $\xi_{air} = \xi_{eq} - \xi_{still}$ , where  $\xi_{still}$  is the identified mechanical damping of the suspension system in Fig. 4. As can be seen, the periodic equivalent damping of linear model compares well with the experimental result in a limited range of amplitudes (around the steady-state amplitude). For amplitudes that are larger than the steady-state amplitude, the simulated results by the linear model are obviously smaller than the experimental results. This is the reason why the simulated transient part of displacement in Fig. 10 decreases slower than the experimental result.

Since the periodic equivalent damping of the linear model is only



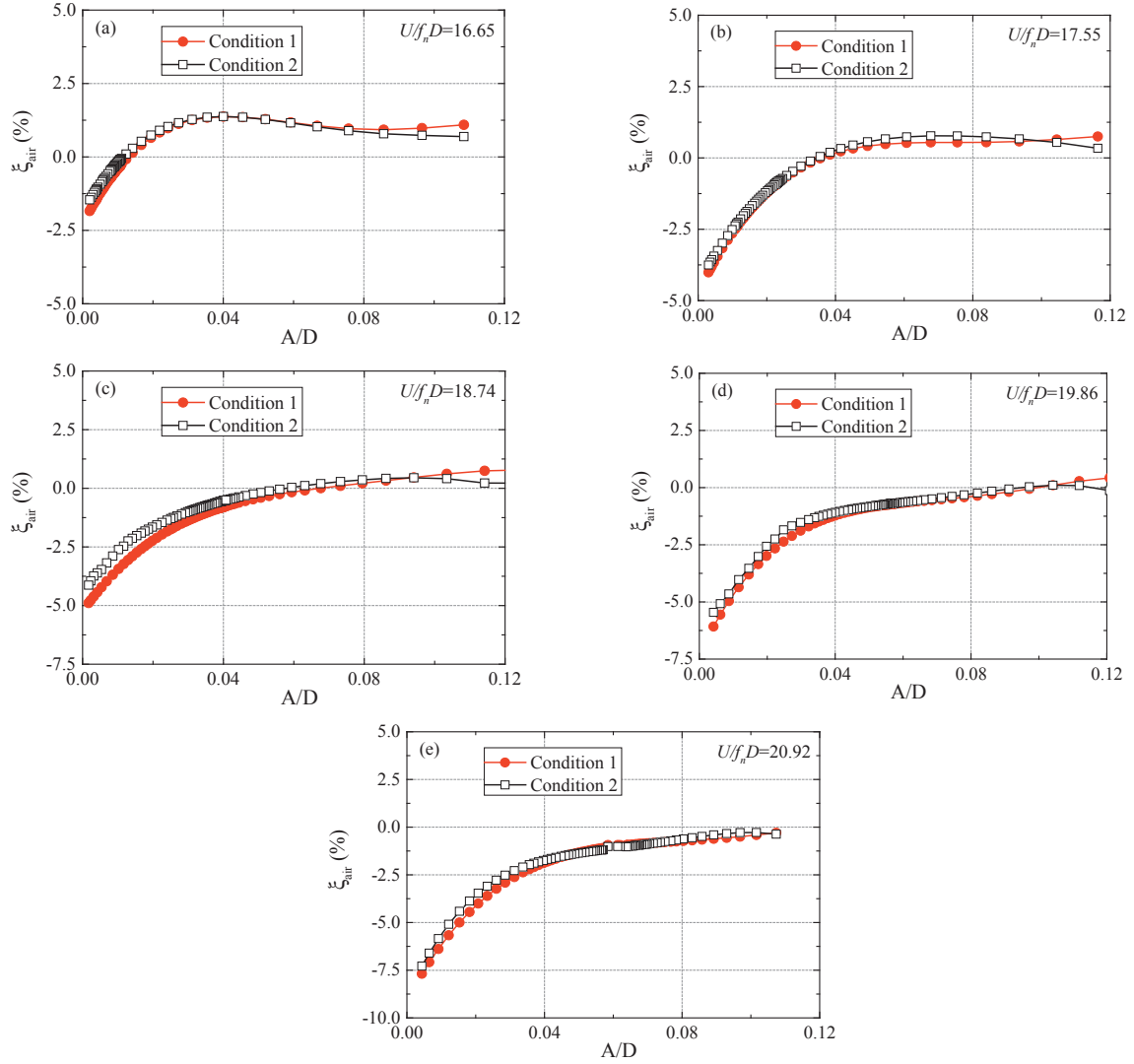


Fig. 9. Additional aeroelastic damping on Condition 1 and Condition (2).

comparable with the experimental result in a limited range of amplitude (around the steady-state amplitude of the deck on Condition (1)), if the Scruton number of the deck changes, the simulated steady-state amplitude of the deck may differ from the experimental result. For validation, the predicted VIV response of the deck on Condition (2) with model parameters identified on Condition (1) is compared with the experimental result, as show in Fig. 12. The predicted non-dimensional steady-state VIV amplitude is 0.048, while the experimental result is 0.057. The prediction error is  $(0.048-0.057)/0.057 = -13.5\%$ .

The above results indicate that the linear empirical model can only reproduce the nonlinear property of the VIV system in a limited range of amplitude (near the steady-state VIV amplitude). For amplitudes out of that amplitude range, the prediction errors are obvious.

#### 4.3. Nonlinear empirical model by Scanlan

The non-dimensional nonlinear empirical model has the following form [8]:

$$Y'' + 2\xi_n K_n Y' + K_n^2 Y = \frac{\rho D^2}{m} [Y_1(K)(1 - \varepsilon Y^2)Y' + Y_2(K)Y + C_L \sin(Ks + \psi)] \quad (24)$$

where  $Y_1$  and  $\varepsilon$  are linear and nonlinear aeroelastic damping parameters;  $Y_2$  is aeroelastic stiffness parameter;  $C_L$  is fluctuating lift coefficient due to vortex shedding;  $\psi$  is phase angle. The other symbols are identical with those defined above.

According to Ehsan and Scanlan [8], the fluctuating lift coefficient  $C_L$  is often neglected. The aeroelastic stiffness  $Y_2$  is determined through shift between structural natural frequency and the frequency of the VIV

Table 3  
Linear empirical model parameters ( $U/f_n D = 19.86$ ).

| Condition (1)<br>$S_c = 24.5$ | $Ks$    | $A_0$  | $A_n$  | $A_s$ | $\xi$  | $\bar{\xi}$ | $\xi_n$ | $H_1^*$ | $C_L$ |
|-------------------------------|---------|--------|--------|-------|--------|-------------|---------|---------|-------|
|                               | $2\pi$  | 0.1773 | 0.1652 | 0.074 | 0.0113 | 0.0098      | 0.0041  | -69.902 | 0.89  |
|                               | $10\pi$ | 0.1773 | 0.1309 | 0.074 | 0.0097 |             |         |         |       |
|                               | $20\pi$ | 0.1773 | 0.1041 | 0.074 | 0.0085 |             |         |         |       |

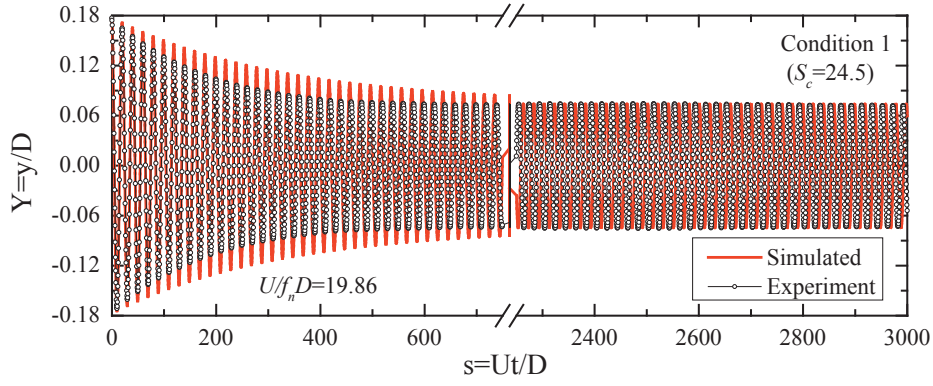


Fig. 10. Comparison between simulated displacement and experimental result.

system:

$$Y_2 = \frac{m}{\rho D^2} (K_n^2 - K^2) \quad (25)$$

The aeroelastic damping  $Y_1$  and  $\varepsilon$  are obtained through fitting the transient displacement envelop of the deck (obtained through decay- or grow-to-resonance):

$$A(s) = \frac{\beta}{\sqrt{1 - \left( \frac{A_0^2 - \beta^2}{A_0^2} \right) e^{-(\alpha\beta^2/4)s}}} \quad (26)$$

in which

$$\alpha = \varepsilon m_r Y_1 \quad (27)$$

$$\beta = \sqrt{(4m_r Y_1 - 8\xi_n K_n)/(\varepsilon m_r Y_1)} \quad (28)$$

where  $A_0$  is the initial point of the envelope;  $A(s)$  is the envelope at non-dimensional time  $s$ ;  $\beta$  is steady-state VIV amplitude.

The parameter  $\alpha$  is firstly obtained through linear regression of the experimental result:

$$Z(s) = \ln \left\{ \frac{A_0^2 [A^2(s) - \beta^2]}{A^2(s)(A_0^2 - \beta^2)} \right\} = -\frac{\alpha\beta^2}{4}s \quad (29)$$

Then,  $Y_1$  and  $\varepsilon$  are determined through Eqs. (27) and (28).

Taking the VIV response of the deck at  $U/f_n D = 19.86$  on Condition (1) as an illustration. The model parameters are identified by using both decay- and grow-to-resonance displacements. According to Marra [20], the values of model parameters are influenced by the length of transient displacement adopted for identification. Thus, several different lengths of transient displacements are adopted, as summarized in Table 4. The mechanical damping of the suspension system  $\xi_n$  adopts the averaged

value of  $\xi_{still}$  in Fig. 4. As can be seen, the identified model parameters vary with the length of transient displacement. Moreover, the model parameters obtained through decay-to-resonance significantly differ from those through grow-to-resonance.

Fig. 13 shows the simulated displacements with the averaged model parameters in Table 4. The simulated decay-to-resonance displacement compares well with the experimental result, as shown in Fig. 13(a). While, the simulated grow-to-resonance displacement varies with the experimental result in the transient part, as shown in Fig. 13(b). This indicates that the model does not well reproduce the nonlinear property of the VIV system in the process of grow-to-resonance.

To investigate the difference between simulated and experimental results, the periodic equivalent damping of the nonlinear empirical model is obtained:

$$\xi_{eq} = \xi_n \frac{K_n}{K} - \frac{m_r Y_1}{2K} + \frac{m_r Y_1 \varepsilon}{8K} \left( \frac{A}{D} \right)^2 \quad (30)$$

Since the damping of suspension system in this study is not constant, to deduct its influence, the aeroelastic damping is calculated through the following equation:

$$\xi_{air} = \xi_{eq} - \xi_{still} \quad (31)$$

in which,  $\xi_{still}$  is mechanical damping of the suspension system in Fig. 4.

Fig. 14 compares the periodic equivalent damping of the nonlinear empirical model and experimental result. As can be seen, the periodic equivalent damping of the model is comparable with the experimental result in a limited range of amplitudes. The simulated damping curve versus structural amplitude has a concave shape, the incremental rate of which is positive (as shown by Eq. (30)). While, the experimental result has a convex shape, the incremental rate of which is negative. Since the model parameters are determined based on least-square

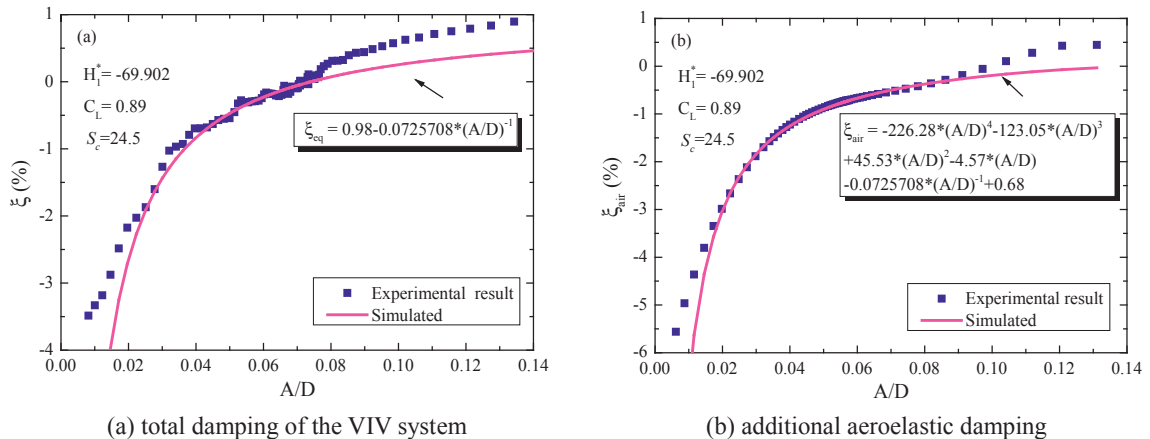


Fig. 11. Comparison between periodic equivalent damping of linear empirical model and experimental results.

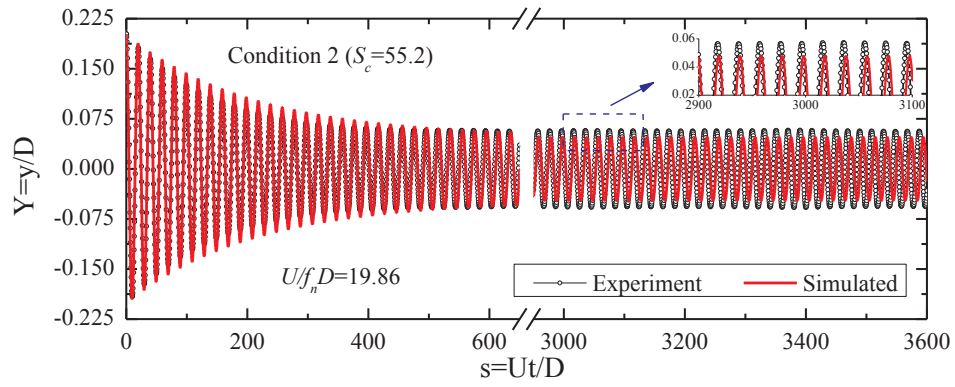


Fig. 12. Predicted VIV response of Condition (2) with Condition (1) parameters.

Table 4

Model parameters of nonlinear empirical model (Condition (1),  $U/f_n D = 19.86$ ).

| Decay-to-resonance | $\Delta s$ | $A_0$   | $\beta$ | $\xi_n$ | $Y_1$ | $\varepsilon$ | $\bar{Y}_1$ | $\bar{\varepsilon}$ | $Y_2$ |
|--------------------|------------|---------|---------|---------|-------|---------------|-------------|---------------------|-------|
|                    | 0–200      | 0.1185  | 0.074   | 0.0041  | 17.75 | 403.13        | 23.73       | 476.18              | 2.78  |
|                    | 0–300      | 0.1185  | 0.074   | 0.0041  | 21.69 | 462.56        |             |                     |       |
|                    | 0–400      | 0.1185  | 0.074   | 0.0041  | 25.40 | 501.68        |             |                     |       |
|                    | 0–500      | 0.1185  | 0.074   | 0.0041  | 30.08 | 537.33        |             |                     |       |
| Grow-to-resonance  | $\Delta s$ | $A_0$   | $\beta$ | $\xi_n$ | $Y_1$ | $\varepsilon$ | $\bar{Y}_1$ | $\bar{\varepsilon}$ | $Y_2$ |
|                    | 0–200      | 0.00083 | 0.074   | 0.0041  | 85.22 | 662.28        | 65.10       | 637.37              | 0.79  |
|                    | 0–400      | 0.00083 | 0.074   | 0.0041  | 68.98 | 646.23        |             |                     |       |
|                    | 0–600      | 0.00083 | 0.074   | 0.0041  | 57.04 | 628.59        |             |                     |       |
|                    | 0–800      | 0.00083 | 0.074   | 0.0041  | 49.19 | 612.36        |             |                     |       |

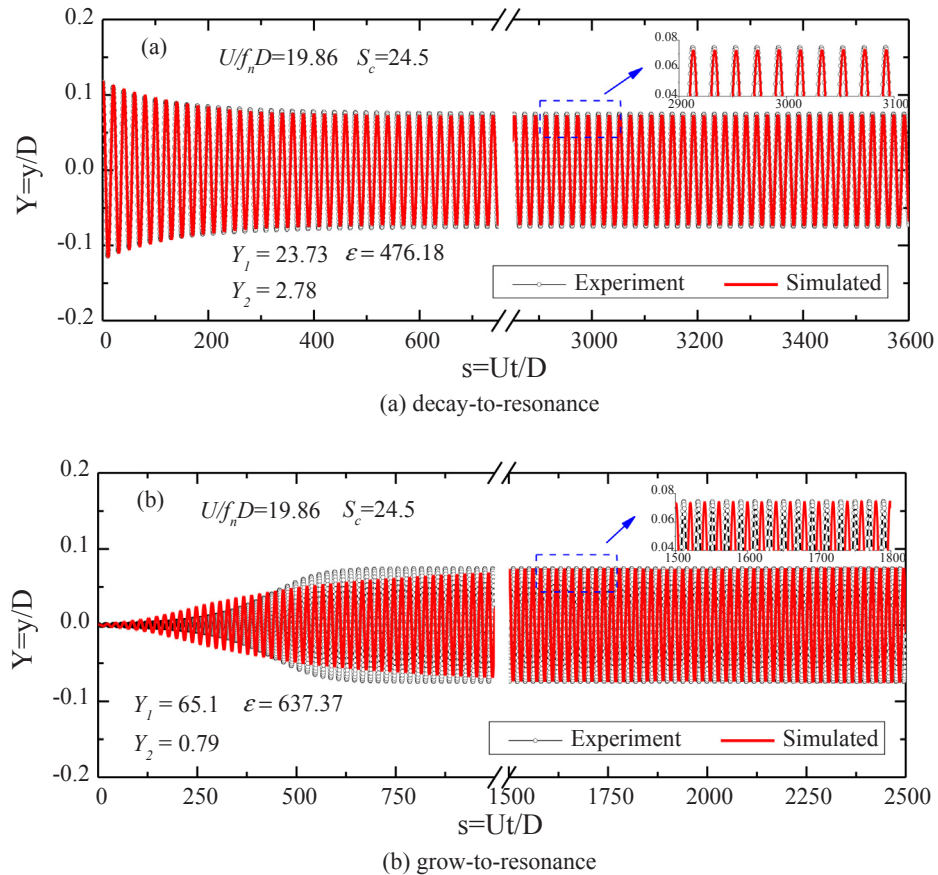
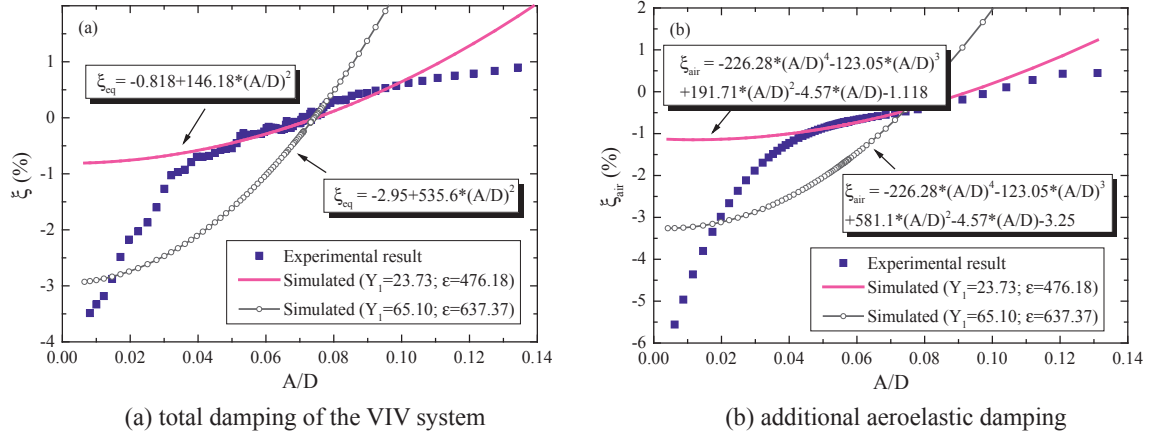
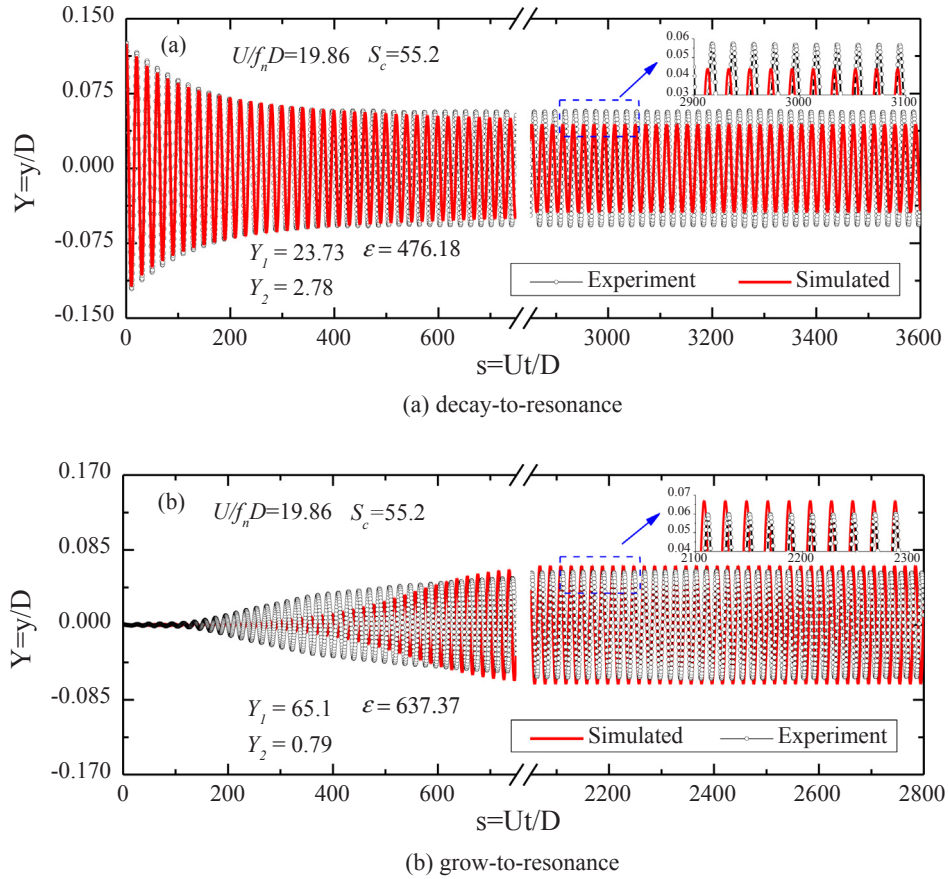


Fig. 13. Simulated displacements by nonlinear empirical model and experimental results (Condition (1)).



**Fig. 14.** Comparison between periodic equivalent damping of nonlinear empirical model and experimental results (squares denote experimental result; solid lines are simulated result with parameters of decay-to-resonance; circle lines are simulated result with parameters of grow-to-resonance).



**Fig. 15.** Predicted displacements by nonlinear empirical model and experimental results (Condition (2)).

fittings, even though the nonlinear property of the model varies with the experimental result, the simulated values with decay-to-resonance parameters are closed to the experimental result in the amplitude range of  $0.7 < A/D < 0.12$ . While, for the simulated result with grow-to-resonance parameters, it significantly differs from the experimental result. This is the reason why the simulated decay-to-resonance displacement is more comparable with the experimental result than grow-to-resonance.

Fig. 15 shows the predicted VIV response of the deck on Condition (2) with model parameters identified on Condition (1). The predicted steady-state VIV amplitude in the process of decay-to-resonance is 0.046, while the experimental result is 0.057. The prediction error is

$(0.048-0.057)/0.057 = -15.8\%$ . The predicted steady-state VIV amplitude in the process of grow-to-resonance is 0.067, the prediction error is  $(0.067-0.057)/0.057 = 17.5\%$ .

Fig. 14 and Eq. (30) indicate that the periodic equivalent damping of the nonlinear empirical model versus structural amplitude differs from the experimental result. This may be due to the neglect of the vortex shedding term  $C_L$ , which limits the flexibility of the model in reproducing the nonlinear property of VIV system. To validate this assumption, the periodic equivalent damping of Eq. (24) is obtained:

$$\xi_{eq} = \xi_n \frac{K_n}{K} - \frac{m_r Y_1}{2K} + \frac{m_r Y_1 \varepsilon}{8K} \left( \frac{A}{D} \right)^2 + \frac{m_r C_L \cos(\psi)}{2K^2} \left( \frac{A}{D} \right)^{-1} \quad (32)$$

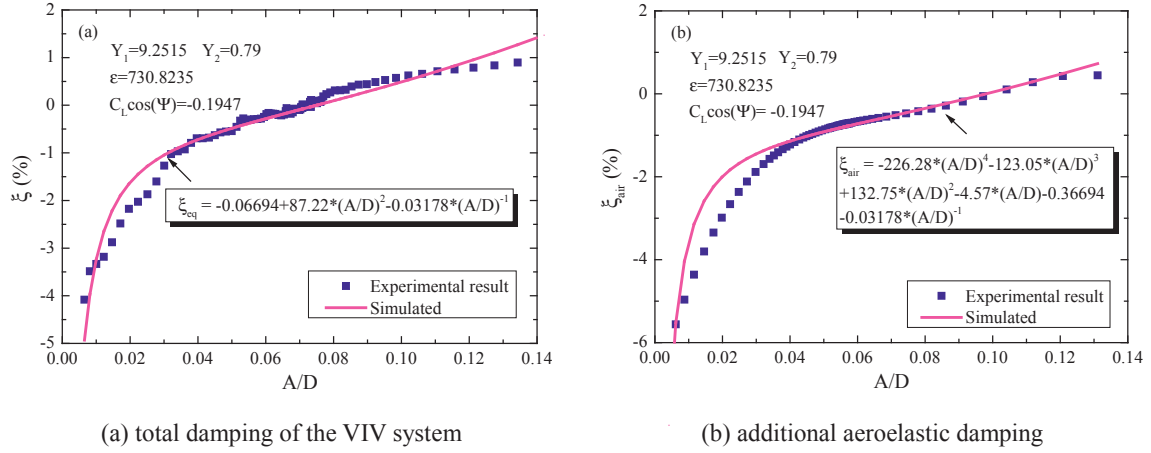


Fig. 16. Directly fitted periodic equivalent damping and experimental result.

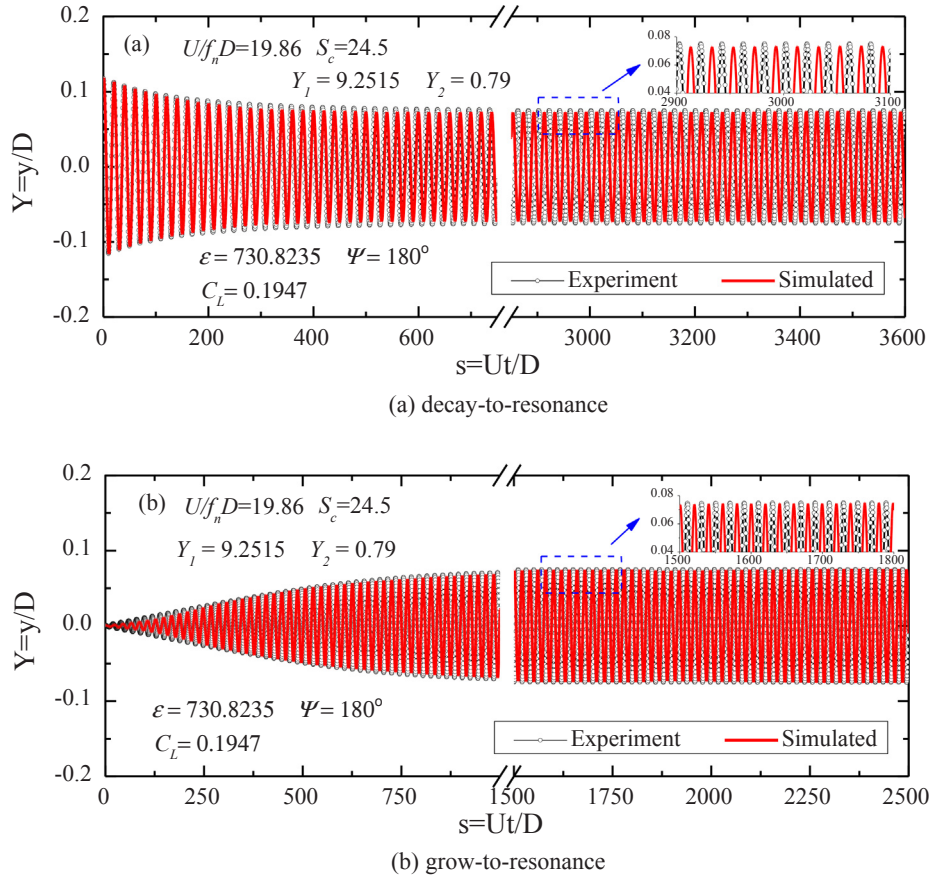


Fig. 17. Simulated displacements by nonlinear empirical model and experimental results (Condition (1)).

The above function is adopted to directly fit the experimental result.

Fig. 16 shows the fitted values of periodic equivalent damping of the nonlinear empirical model. The simulated curve has a convex shape versus structural amplitude, which is identical with the experimental result. Comparing with the result in Fig. 14, the simulated results are also more closed to the experimental results in this figure.

The directly fitted model parameters in Fig. 16 are adopted to simulate the VIV responses of the deck on Condition (1) and Condition (2). It is noticeable that in the periodic equivalent damping of nonlinear empirical model, the energy transferred from flow field to the deck due to the vortex shedding effect is a combination of fluctuating lift coefficient  $C_L$  and phase angle  $\psi$ . Since the values of  $\psi$  and  $C_L$  are unknown,

in the process of calculation, the phase angle is assumed to be  $\psi = 180^\circ$ . This assumption may cause a phase shift in the steady-state displacement, but it does not influence the calculated displacement magnitude.

Fig. 17 shows the simulated VIV response of the deck on Condition (1). Comparing with Fig. 13, the transient part of displacement is more closed to the experimental result, especially for grow-to-resonance. Fig. 18 shows the predicted results on Condition (2). The predicted steady-state VIV amplitude of decay-to-resonance is 0.0495. The prediction error is  $(0.0495-0.057)/0.057 = -13.15\%$ . The predicted steady-state VIV amplitude of grow-to-resonance is 0.048, the prediction error is  $(0.048-0.057)/0.057 = -15.7\%$ . Comparing with Fig. 15, the prediction error is reduced.



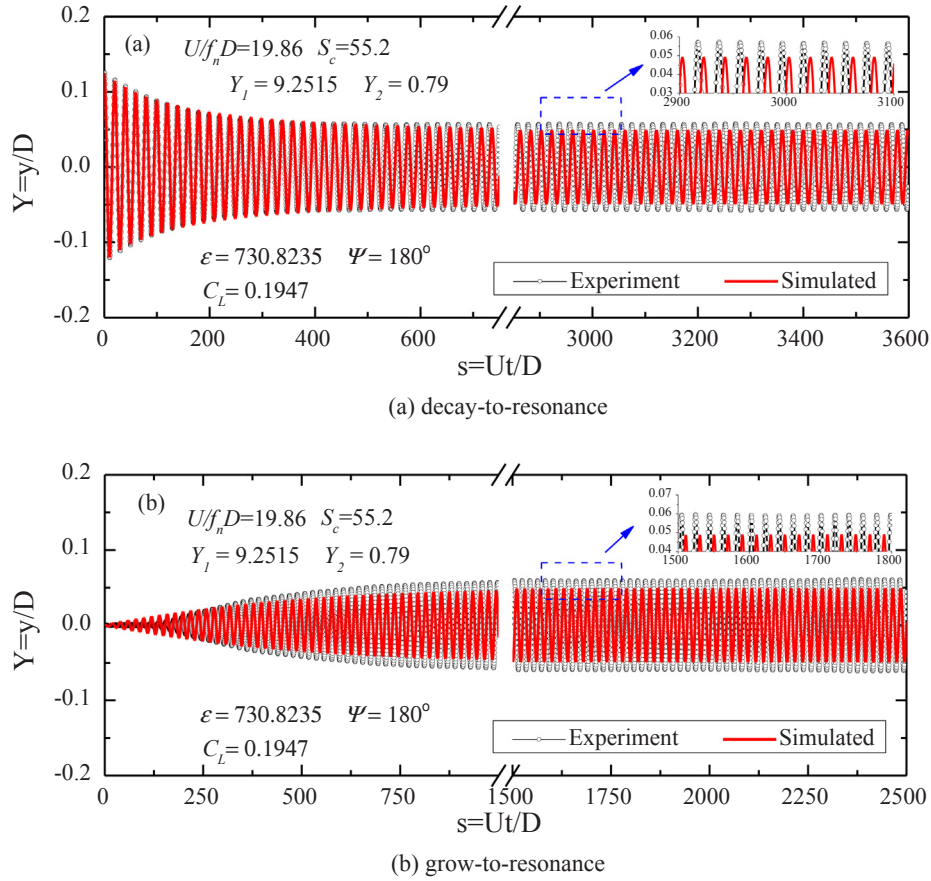


Fig. 18. Predicted displacements by nonlinear empirical model and experimental results (Condition (2)).

Table 5

Model parameters of generalized empirical model (Condition (1),  $U/f_n D = 19.86$ ).

| Decay-to-resonance | $\Delta s$ | $A_0$   | $A_s$ | $\xi_n$ | $Y_1$  | $\varepsilon$ | $\nu$   | $\bar{Y}_1$ | $\bar{\varepsilon}$ | $\bar{\nu}$ |
|--------------------|------------|---------|-------|---------|--------|---------------|---------|-------------|---------------------|-------------|
|                    | 0–250      | 0.1773  | 0.074 | 0.0041  | 16,265 | 1.0058        | 8.2e-4  | 19,587      | 1.00525             | 0.000738    |
|                    | 0–500      | 0.1773  | 0.074 | 0.0041  | 22,808 | 1.0045        | 6.31e-4 |             |                     |             |
|                    | 0–750      | 0.1773  | 0.074 | 0.0041  | 19,142 | 1.0055        | 7.68e-4 |             |                     |             |
|                    | 0–1000     | 0.1773  | 0.074 | 0.0041  | 20,133 | 1.0052        | 7.33e-4 |             |                     |             |
| Grow-to-resonance  | $\Delta s$ | $A_0$   | $A_s$ | $\xi_n$ | $Y_1$  | $\varepsilon$ | $\nu$   | $\bar{Y}_1$ | $\bar{\varepsilon}$ | $\bar{\nu}$ |
|                    | 0–250      | 0.00083 | 0.074 | 0.0041  | 13,007 | 1.0093        | 0.0012  | 32,688      | 1.004               | 0.00056     |
|                    | 0–500      | 0.00083 | 0.074 | 0.0041  | 50,136 | 1.0022        | 0.0003  |             |                     |             |
|                    | 0–750      | 0.00083 | 0.074 | 0.0041  | 30,606 | 1.0034        | 0.0005  |             |                     |             |
|                    | 0–1000     | 0.00083 | 0.074 | 0.0041  | 37,157 | 1.0027        | 0.0004  |             |                     |             |
|                    | 0–1250     | 0.00083 | 0.074 | 0.0041  | 32,534 | 1.0031        | 0.0004  |             |                     |             |

The above results indicate that the neglect of vortex shedding term in the nonlinear empirical model may limit its flexibility in reproducing the nonlinear property of the VIV system. To better simulate the VIV system, the vortex shedding term should be considered. The model parameters can be obtained through directly fitting the periodic equivalent damping of the model with experimental result.

#### 4.4. Generalized empirical model by Larsen

The generalized empirical model has the following form [18]:

$$Y'' + 2\xi_n K_n Y' + K_n^2 Y = m_r Y_1(K) [1 - \varepsilon |Y(s)|^{2\nu}] Y'(s) \quad (33)$$

where  $Y_1$ ,  $\varepsilon$  and  $\nu$  are parameters to be determined through experiment. The other symbols are identical with those defined above.

According to Larsen [18], the parameters  $Y_1$  and  $\nu$  can be obtained through fitting the displacement envelope of decay- or grow-to-resonance:

$$A(s) = \frac{A_s}{\left\{ 1 - \left[ 1 - \left( \frac{A_s}{A_0} \right)^{2\nu} \right] \exp[-\nu(m_r Y_1 - 2K_n \xi_n)s] \right\}^{1/2\nu}} \quad (34)$$

in which,  $A_0$  is initial point of the envelope;  $A(s)$  is the envelope value at non-dimensional time  $s$ ;  $A_s$  is steady-state VIV amplitude. After obtaining  $Y_1$  and  $\nu$ , parameter  $\varepsilon$  can be obtained through solving the following equation:

$$\varepsilon = \frac{Y_1 - 2\xi_n K_n / m_r}{Y_1 I_m A_s^{2\nu}} \quad (35)$$

where

$$I_m = \frac{K_n}{\pi} \int_0^{2\pi/K} |\cos(Ks)|^{2\nu} \cdot \sin^2(Ks) ds \quad (36)$$

Taking the VIV of the deck at  $U/f_n D = 19.86$  on Condition (1) as an illustration, the model parameters are identified by using both decay- and grow-to-resonance displacements, as summarized in Table 5.



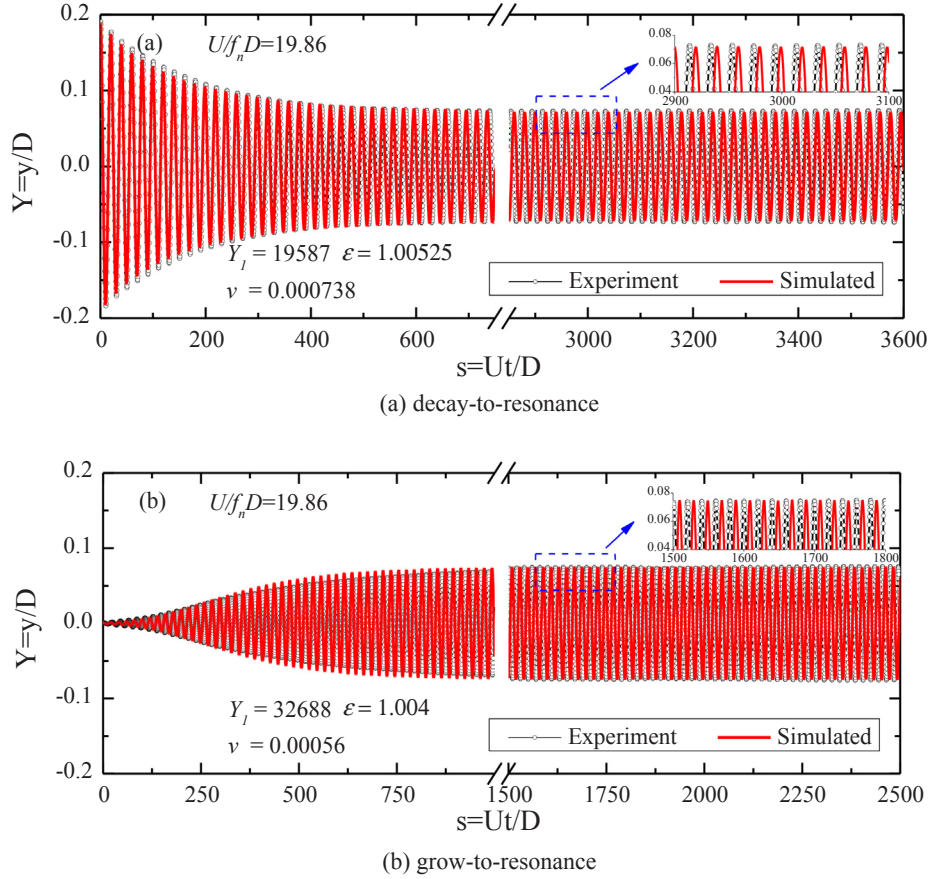


Fig. 19. Simulated displacements by generalized empirical model and experimental results (Condition (1)).

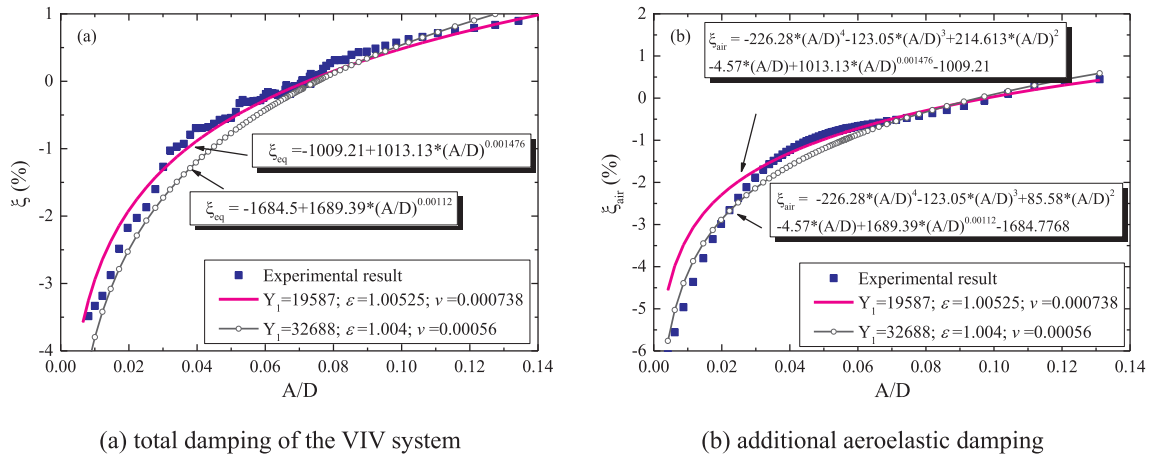


Fig. 20. Periodic equivalent damping of the generalized empirical model and experimental results (squares denote experimental result; solid lines are simulated result with parameters of decay-to-resonance; circle lines are simulated result with parameters of grow-to-resonance).

Fig. 19 shows the simulated displacements and the experimental results. Both the simulated decay- and grow-to-resonances compare well with the experimental results. Comparing with the nonlinear empirical model in Fig. 13, the generalized empirical model performs better in reproducing the grow-to-resonance process. This is because the nonlinear order term in the generalized empirical model (i.e.  $\nu$ ) can be adjusted based on the nonlinear property of experimental result. In this sense, the generalized empirical model is more flexible in reproducing the nonlinearity of the VIV system.

The periodic equivalent damping of the generalized empirical model is also obtained. Due to the nonlinear term  $\nu$ , it is hard to get an

analytical expression for  $\xi_{eq}$ . But after substituting the values of  $\xi_n$ ,  $Y_l$ ,  $\varepsilon$  and  $\nu$  into Eq. (16), the expression of  $\xi_{eq}$  can be obtained through symbolic calculations. With decay-to-resonance parameters, the following equation is obtained:

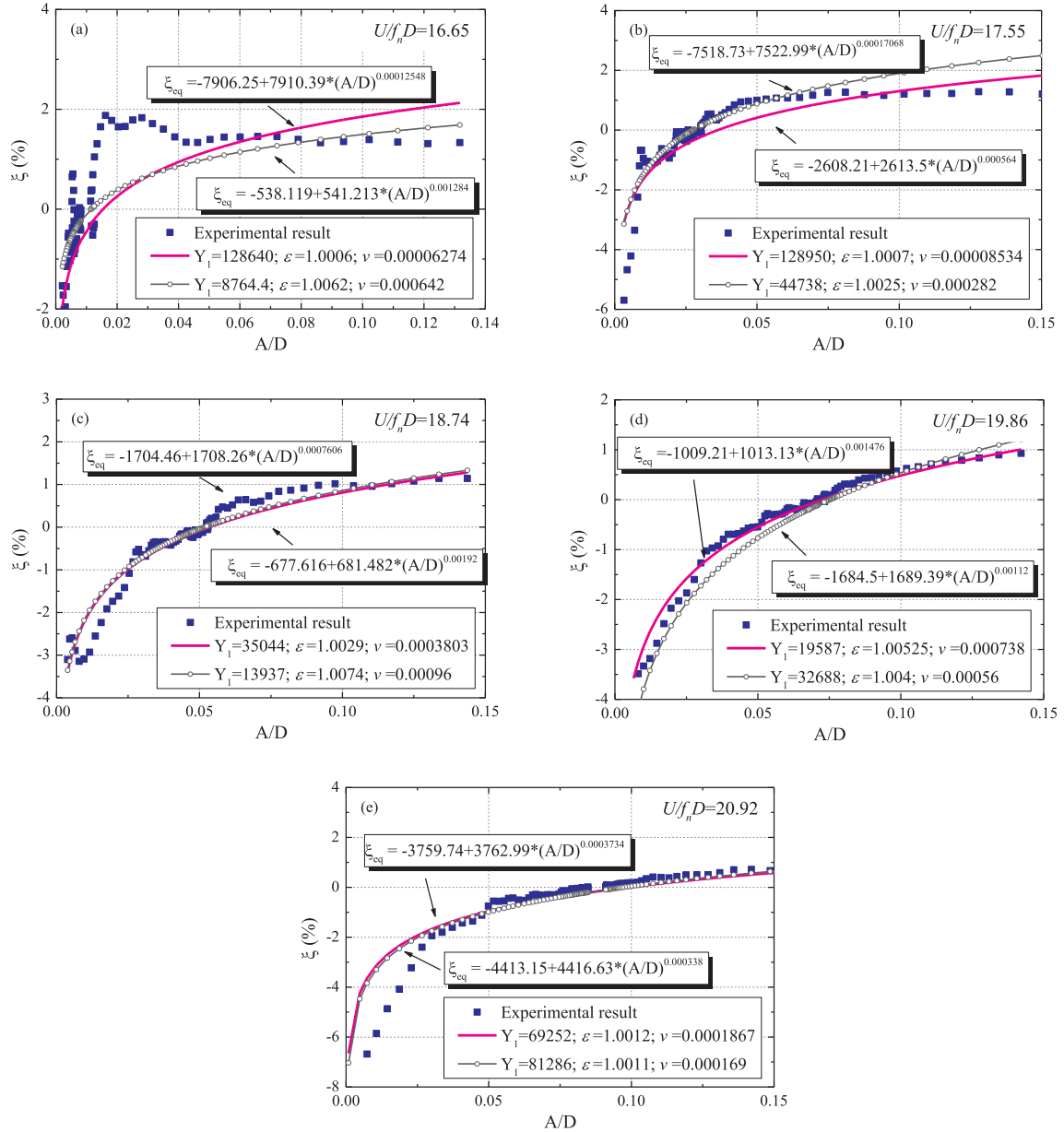
$$\xi_{eq} = -10.0921 + 10.1313 \left( \frac{A}{D} \right)^{0.001476} \quad (37)$$

with grow-to-resonance parameters, the following equation is obtained:

$$\xi_{eq} = -16.845 + 16.894 \left( \frac{A}{D} \right)^{0.00112} \quad (38)$$

**Table 6**  
Model parameters of generalized empirical model at different wind velocities on Condition (1).

| Decay-to-resonance | $U/f_n D$ | $A_0$    | $A_s$ | $\xi_n$ | $Y_1$   | $\varepsilon$ | $\nu$      |
|--------------------|-----------|----------|-------|---------|---------|---------------|------------|
|                    | 16.65     | 0.1768   | 0.012 | 0.0041  | 128,640 | 1.0006        | 0.00006274 |
|                    | 17.55     | 0.184    | 0.03  | 0.0041  | 128,950 | 1.0007        | 0.00008534 |
|                    | 18.74     | 0.1851   | 0.052 | 0.0041  | 35,044  | 1.0029        | 0.0003803  |
|                    | 19.86     | 0.1773   | 0.074 | 0.0041  | 19,587  | 1.00525       | 0.000738   |
|                    | 20.92     | 0.2065   | 0.09  | 0.0041  | 69,252  | 1.0012        | 0.0001867  |
| Grow-to-resonance  | $U/f_n D$ | $A_0$    | $A_s$ | $\xi_n$ | $Y_1$   | $\varepsilon$ | $\nu$      |
|                    | 16.65     | 0.00059  | 0.012 | 0.0041  | 8764.4  | 1.0062        | 0.000642   |
|                    | 17.55     | 0.0016   | 0.03  | 0.0041  | 44,738  | 1.0025        | 0.000282   |
|                    | 18.74     | 0.00011  | 0.052 | 0.0041  | 13,937  | 1.0074        | 0.00096    |
|                    | 19.86     | 0.00083  | 0.074 | 0.0041  | 32,688  | 1.004         | 0.00056    |
|                    | 20.92     | 0.000458 | 0.09  | 0.0041  | 81,286  | 1.0011        | 0.000169   |



**Fig. 21.** Simulated total damping by generalized empirical model and experimental results on Condition (1) (squares denote experimental result; solid lines are simulated result with parameters of decay-to-resonance; circle lines are simulated result with parameters of grow-to-resonance).

Fig. 20 shows the periodic equivalent damping of the generalized empirical model and the experimental result. The generalized empirical model compares well with the experimental result. As discussed above, this is because the nonlinear order term  $\nu$  is adjustable, which makes

this model more flexible in reproducing the nonlinear property of the VIV system.

To further evaluate the capacity of the generalized empirical model, the model parameters at different wind velocities on Condition (1) are

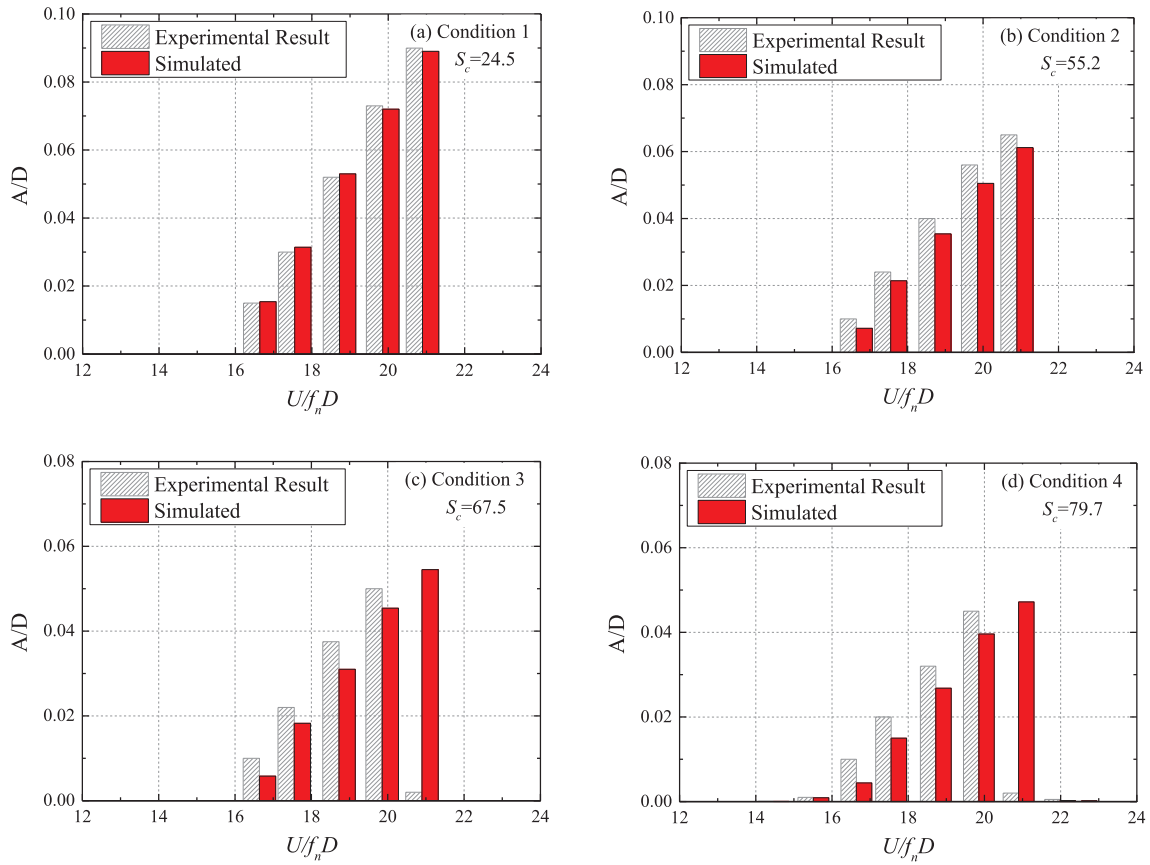


Fig. 22. Predicted VIV amplitudes at different damping conditions by generalized empirical model.

obtained, as summarized in Table 6. The periodic equivalent damping of the generalized empirical model at different wind velocities are compared with experimental results, as shown in Fig. 21.

As can be seen, the generalized empirical model can reproduce the convex curve shape of the total damping versus structural amplitude. However, the simulated damping curves are not always identical with the experimental results. For example, the simulated damping curve in Fig. 21(a) obviously differs from the experimental result in the amplitude range of  $0.01 < A/D < 0.04$ , and the curves in Fig. 21(e) also differ from the experimental result in the amplitude range of  $0 < A/D < 0.03$ . This is because, on one hand, the nonlinear property of the generalized empirical model is determined by a single parameter (i.e. the nonlinear order term  $\nu$ ), which limits the feasibility of this model once the damping curve versus structural amplitude varies with the smooth convex type. On the other hand, the model parameters of the generalized empirical model are determined through least-square fittings of the displacement envelope. The damping ratio is a logarithmic function of the displacement envelope, and a small fitting error in displacement envelope can result in an obvious deviation in the damping of the system.

Fig. 22 compares the predicted steady-state VIV amplitudes of the deck at different damping conditions with model parameters of Condition (1) (Table 6). The simulated results compare well with the experiment on Condition (1) (Fig. 22(a)). While, for the other three damping conditions, the predicted amplitudes are slightly lower than the experimental results. The prediction error is around 10%. This is because, the calculated steady-state VIV amplitudes depend on the precision of aeroelastic damping curves within the amplitude range where the total damping is near zero. For Condition (1), the damping curves are almost identical with experimental results within that range, the prediction error is thus quite small. While, for Conditions 2–4, the mechanical damping increases, the predicted results thus depend on the

precision of damping curves lower than that range, where the damping curves slightly vary with the experimental results. Thus, the predicted errors are more obvious than Condition (1).

It is noticeable that the increase of Scruton number leads to the sudden change of stability of the VIV system at  $U/f_n D = 20.92$  for Condition (3) and Condition (4) (the VIV disappears). The predicted errors are thus quite obvious at this wind velocity. This phenomenon cannot be simulated by using model parameters identified on steady VIV systems.

#### 4.5. Generalized polynomial model

Mashnad and Jones [22] proposed to directly fit the experimentally identified aeroelastic damping by using polynomial function. The steady-state VIV amplitudes were calculated through an iteration procedure. Later, Xu et al [21] proposed to fit the experimentally identified aeroelastic damping through a generalized polynomial model. The harmonic balance method was adopted to transform the function of structural amplitude to that of structural velocity. The LCO displacements of the deck were then calculated through numerical integration method.

The generalized polynomial model can be expressed as:

$$\begin{aligned}
 & Y'' + 2\zeta_n K_n Y' + K_n^2 Y \\
 &= m_r \left[ \sum_{i=1}^m \frac{A_i}{K^{i-1}} (Y')^{i-1} + \sum_{j=2}^n \frac{A_j}{K^{j-1}} |Y'|^{j-1} \right] Y' + m_r C_L \sin(Ks + \psi)
 \end{aligned} \quad (39)$$

where  $A_i$  ( $i$  is an odd number) and  $A_j$  ( $j$  is an even number) are parameters to be determined through experiment.

Taking the 5th-order polynomial function as an illustration, the periodic equivalent damping of the model is:

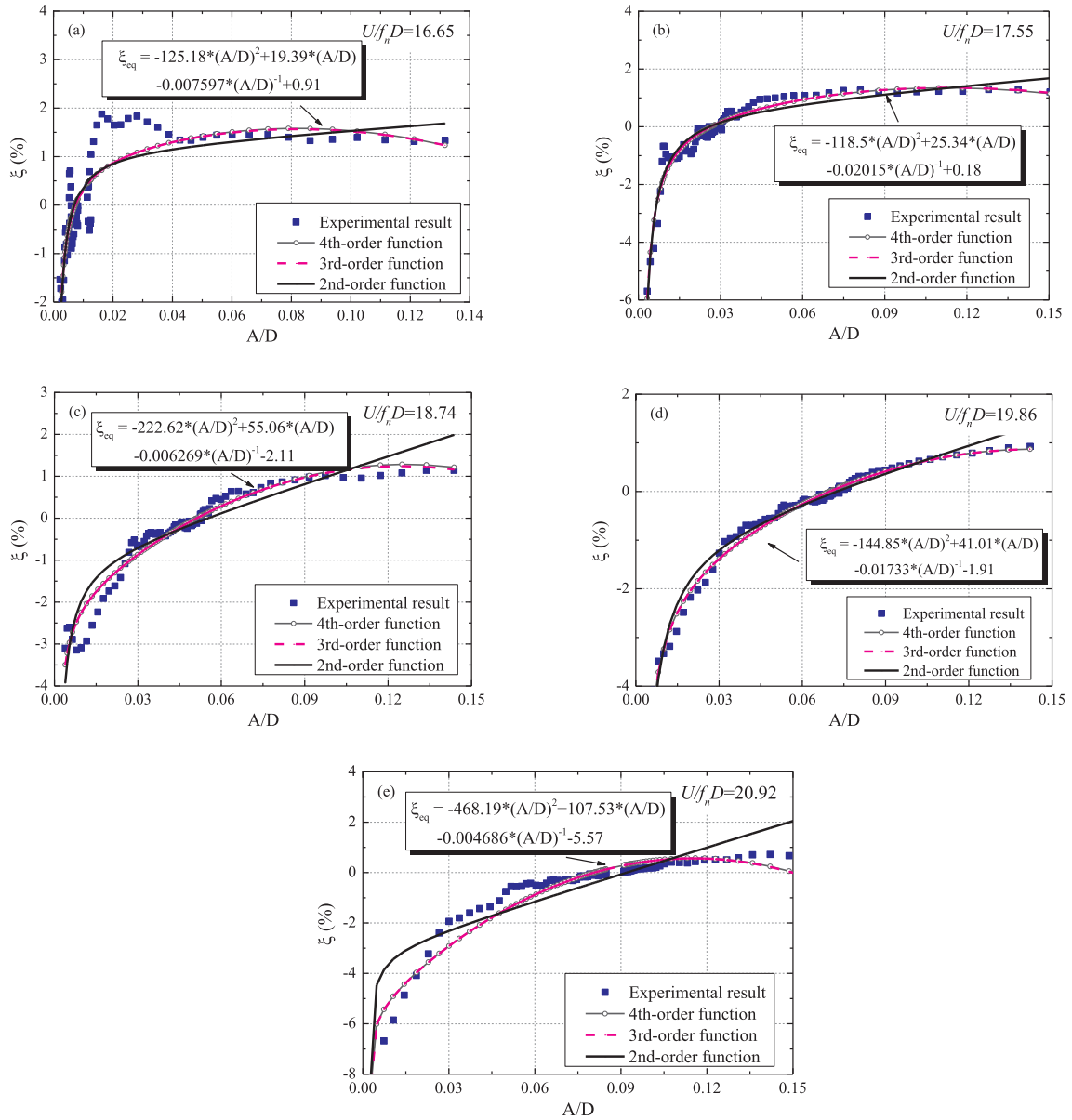


Fig. 23. Simulated total damping by generalized polynomial model and experimental results on Condition (1) (squares denote experimental result; solid, dashed and circle lines are polynomial fitting with different orders).

Table 7

Model parameters of generalized polynomial model.

| $U/f_n D$ | $A_3$    | $A_2$    | $A_1$  | $C_L \cos(\psi)$ |
|-----------|----------|----------|--------|------------------|
| 16.65     | 3861.23  | -528.59  | -11.67 | -0.066           |
| 17.55     | 3467.65  | -655.21  | 5.16   | -0.1583          |
| 18.74     | 6101.46  | -1333.45 | 51.73  | -0.0432          |
| 19.86     | 3746.56  | -937.12  | 44.92  | -0.1064          |
| 20.92     | 11497.34 | -2333.17 | 110.05 | -0.0259          |

$$\xi_{eq} = \frac{\xi_n K_n}{K} - \frac{5A_3 m_r}{16K} \left(\frac{A}{D}\right)^4 - \frac{16A_4 m_r}{15K\pi} \left(\frac{A}{D}\right)^3 - \frac{3A_5 m_r}{8K} \left(\frac{A}{D}\right)^2 - \frac{4A_2 m_r}{3K\pi} \left(\frac{A}{D}\right) - \frac{A_1 m_r}{2K} + \frac{m_r C_L \cos(\psi)}{2K^2} \left(\frac{A}{D}\right)^{-1} \quad (40)$$

The periodic equivalent damping of the model can be used to directly fit the experimentally identified damping of the VIV system.

Fig. 23 shows the fitted values of the generalized polynomial model with different orders of nonlinearity. As can be seen, the 3rd-order polynomial function is adequate to reproduce the nonlinear damping of the VIV system, an increase of nonlinear order does not obviously

improve the fitting performance of this model. The model parameters of the 3rd-order polynomial function are summarized in Table 7.

Comparing with generalized empirical model in Fig. 21, the residual errors by this model are further reduced, such as in the amplitude range of  $0 < A/D < 0.03$  at  $U/f_n D = 20.92$ . However, the simulated results are still not exactly identical with the experimental results, such as in the range of  $0.01 < A/D < 0.04$  at  $U/f_n D = 16.65$ . This is because, on one hand, only 3rd-order polynomial function is adopted. On the other hand, the amplitude is normalized with deck height. Since the value of non-dimensional amplitude is much smaller than unit, the higher-order polynomial terms are less sensitivity in the process of least square fitting. This is the reason why the coefficients of higher-order terms in Table 7 are much larger than those of lower-order terms and why the involving of 4th-order polynomial does not obviously improve the fitting performance. This situation can be changed if the physical quantities of amplitude are taken for polynomial fitting.

Fig. 24 shows the predicted VIV amplitudes of the deck at different damping conditions with parameters of Condition (1). The results are obtained through numerical integration of Eq. (39). The phase angle is

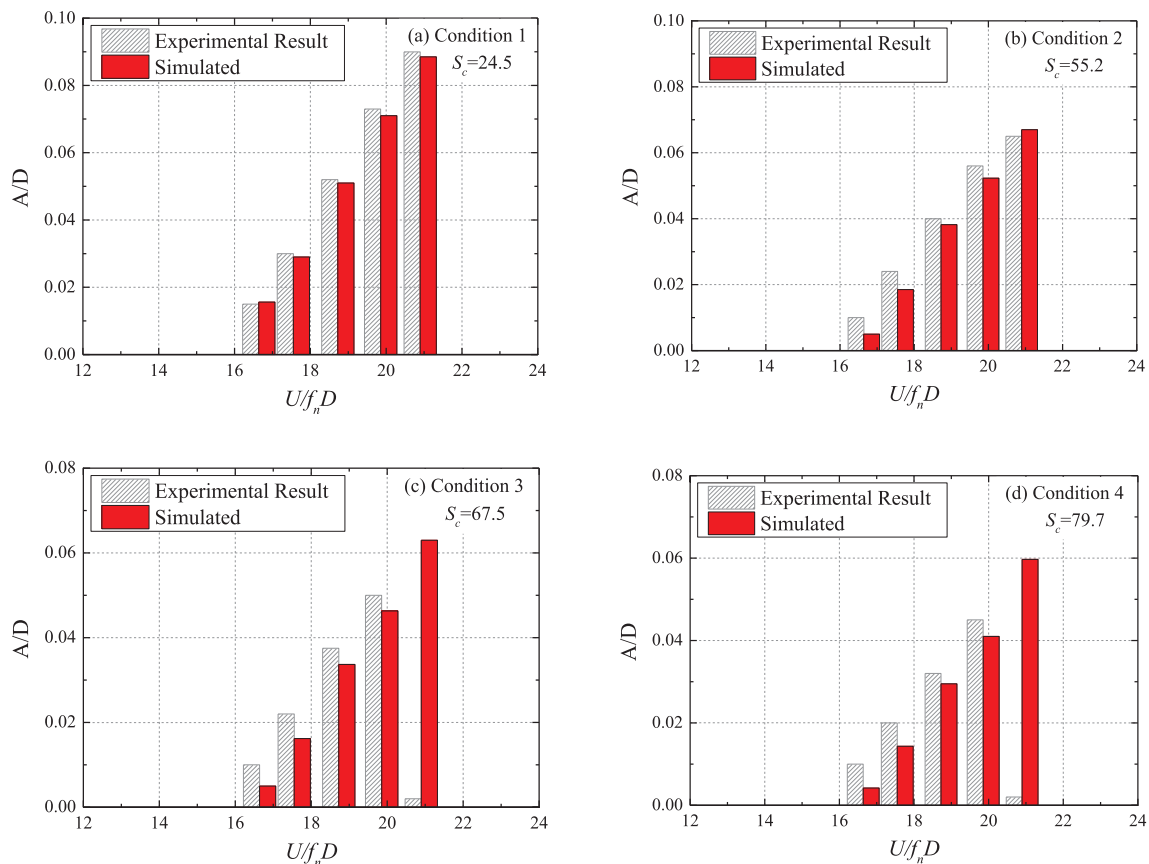


Fig. 24. Predicted VIV amplitudes at different damping conditions by generalized polynomial model.

assumed to be  $\psi = 180^\circ$ . As discussed above, this assumption will not influence the calculated magnitude of displacement. Comparing with the generalized empirical model in Fig. 23, the predicted errors on Conditions 2–4 are slightly reduced. The results further validate that by directly fitting the aeroelastic damping of the VIV system, the prediction errors can be reduced once the Scruton number of the bridge varies with the one on which the model parameters were identified.

Similar with the generalized empirical model, the predicted results at  $U/f_n D = 20.92$  for Condition (3) and Condition (4) obviously differ from the experiment. As discussed above, this is because the increase of Scruton number leads to the sudden change of stability of the VIV system, and this phenomenon cannot be reproduced with model parameters identified on steady VIV systems.

## 5. Conclusions

Wind tunnel experiment was conducted for evaluating the capabilities of some commonly used bridge VIV empirical models. An instantaneous identification method is adopted for obtaining the properties of aeroelastic damping and stiffness of the VIV system. The simulated aeroelastic damping by the empirical models as well as the predicted VIV responses of the deck at different Scruton numbers are compared with experimental results. The main results are as follows:

- (1) The damping of the VIV system shows a nonlinear characteristic versus structural amplitude, whereas the variation of frequency versus structural amplitude is quite small, which can be taken as constant.
- (2) The more accurate the simulated aeroelastic damping by an empirical model, the smaller the prediction errors by this model will be. Through directly fitting the aeroelastic damping, the prediction errors by an empirical model can be further reduced.

- (3) The increase of Scruton number leads to the sudden change of stability of the VIV system (the VIV disappears) at the end part of lock-in wind velocity region. This phenomenon cannot be predicted by using empirical models with parameters identified on steady VIV systems.
- (4) The results are based on a single deck shape and only the most commonly used SDOF VIV empirical models are involved. Future studies on different deck shapes and/or empirical models can be conducted by using the procedures in this study.

## CRediT authorship contribution statement

**Kun Xu:** Investigation. **Yaojun Ge:** Resources, Supervision, Project administration. **Lin Zhao:** Methodology, Validation.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.engstruct.2020.110860>.



## References

- [1] Li H, Laima S, Zhang Q, Li N, Liu Z. Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge. *J Wind Eng Ind Aerodyn* 2014;124:54–67.
- [2] Li H, Laima S, Ou J, Zhao X, Zhou W, Yu Y, et al. Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements. *Eng Struct* 2011;33:1894–907.
- [3] Fujino Y, Yoshida Y. Wind-induced vibration and control of Trans-Tokyo Bay Crossing Bridge. *J Struct Eng* 2002;128:1012–25.
- [4] Larsen A, Esdahl S, Andersen JE, Vejrum T. Storebælt suspension bridge – vortex shedding excitation and mitigation by guide vanes. *J Wind Eng Ind Aerodyn* 2000;88:283–96.
- [5] Li H, Laima S, Jing H. Reynolds number effects on aerodynamic characteristics and vortex-induced vibration of a twin-box girder. *J Fluids Struct* 2014;50:358–75.
- [6] Laima S, Jiang C, Li H, Chen W, Ou J. A numerical investigation of Reynolds number sensitivity of flow characteristics around a twin-box girder. *J Wind Eng Ind Aerodyn* 2018;172:298–316.
- [7] Frandsen JB. Simultaneous pressures and accelerations measured full-scale on the Great Belt East suspension bridge. *J Wind Eng Ind Aerodyn* 2001;89:95–129.
- [8] Ehsan F, Scanlan RH. Vortex-induced vibrations of flexible bridges. *J Eng Mech* 1990;116:1392–411.
- [9] Sun Y, Li M, Liao H. Nonlinear approach of vortex-induced vibration for line-like structures. *J Wind Eng Ind Aerodyn* 2014;124:1–6.
- [10] Barhoush H, Namini AH, Skop RA. Vortex shedding analysis by finite elements. *J Sound Vib* 1995;184:111–27.
- [11] Lewandowski R. Application of semi-empirical model to analysis of vortex-excited vibrations of beams near synchronisation region. Civil-Comp press; 2000.
- [12] Xu K, Ge Y, Zhao L, Du X. Experimental and numerical study on the dynamic stability of vortex-induced vibration of bridge decks. *Int J Struct Stab Dyn* 2018;18:1850033.
- [13] Williamson CHK, Govardhan R. Vortex-induced vibrations. *Annu Rev Fluid Mech* 2004;36:413–55.
- [14] Sarpkaya T, Sarpkaya T. A critical review of the intrinsic nature of vortex-induced vibrations. *J Fluids Struct* 2004;19:389–447.
- [15] Bishop RED, Hassan AY. The lift and drag forces on a circular cylinder in a flowing fluid. *Royal Soc London Proc* 1964;277:32–50.
- [16] Scanlan RH. State-of-the-art methods for calculating flutter, vortex-induced, and buffeting response of bridge structures; 1981.
- [17] Scanlan RH. Bridge flutter derivatives at vortex lock-in. *J Struct Eng* 1998;124:450–8.
- [18] Larsen A. A generalized model for assessment of vortex-induced vibrations of flexible structures. *J Wind Eng Ind Aerodyn* 1995;57:281–94.
- [19] Marra AM, Mannini C, Bartoli G. Measurements and improved model of vortex-induced vibration for an elongated rectangular cylinder. *J Wind Eng Ind Aerodyn* 2015;147:358–67.
- [20] Marra AM, Mannini C, Bartoli G. Van der Pol-type equation for modeling vortex-induced oscillations of bridge decks. *J Wind Eng Ind Aerodyn* 2011;99:776–85.
- [21] Xu K, Ge Y, Zhao L, Du X. Calculating vortex-induced vibration of bridge decks at different mass-damping conditions. *J Bridge Eng* 2018;23:04017149.
- [22] Mashnad M, Jones NP. A model for vortex-induced vibration analysis of long-span bridges. *J Wind Eng Ind Aerodyn* 2014;134:96–108.
- [23] Diana G, Resta F, Belloli M, Rocchi D. On the vortex shedding forcing on suspension bridge deck. *J Wind Eng Ind Aerodyn* 2006;94:341–63.
- [24] Zhu LD, Meng XL, Guo ZS. Nonlinear mathematical model of vortex-induced vertical force on a flat closed-box bridge deck. *J Wind Eng Ind Aerodyn* 2013;122:69–82.
- [25] Wu T, Kareem A. Vortex-induced vibration of bridge decks: Volterra series-based model. *J Eng Mech* 2013;139:1831–43.
- [26] Xu K, Zhao L, Ge Y. Reduced-order modeling and calculation of vortex-induced vibration for large-span bridges. *J Wind Eng Ind Aerodyn* 2017;167:228–41.
- [27] Li S, Kaiser E, Laima S, Li H, Brunton SL, Kutz JN. Discovering time-varying aerodynamics of a prototype bridge by sparse identification of nonlinear dynamical systems. *Phys Rev E* 2019;100:022220.
- [28] Li S, Laima S, Li H. Data-driven modeling of vortex-induced vibration of a long-span suspension bridge using decision tree learning and support vector regression. *J Wind Eng Ind Aerodyn* 2018;172:196–211.
- [29] Zhang Z, Ge Y, Chen Z. Vortex-induced oscillations of bridges: theoretical linkages between sectional model tests and full bridge responses. *Wind Struct* 2014;19:233–47.
- [30] Owen JS, Vann AM, Davies JP, Blakeborough A. The prototype testing of Kessock Bridge: response to vortex shedding. *J Wind Eng Ind Aerodyn* 1996;60:91–108.
- [31] Macdonald JHG, Irwin PA, Fletcher MS. Vortex-induced vibrations of the Second Severn Crossing cable-stayed bridge - Full-scale and wind tunnel measurements. *Proc Inst Civil Eng Struct Build* 2002;152:123–34.
- [32] Burden AR. Japanese cable-stayed bridge design. *Proc Inst Civil Eng Part Des Constr* 1991;90:1021–51.
- [33] Seo JW, Kim HK, Jin P, Kim KT, Kim GN. Interference effect on vortex-induced vibration in a parallel twin cable-stayed bridge. *J Wind Eng Ind Aerodyn* 2013;116:7–20.
- [34] Smith IJ. Wind induced dynamic response of the Wye Bridge. *Eng Struct* 1980;2:202–8.
- [35] Ozkan E, Main J, Jones NP. Full scale measurements of the Fred Hartman Bridge. *JWE* 2001;89.
- [36] Larsen A, Esdahl S, Andersen JE, Vejrum T. Storebælt suspension bridge–vortex shedding excitation and mitigation by guide vanes. *J Wind Eng Ind Aerodyn* 2000;88:283–96.
- [37] Matsumoto M, Shirato H, Yagi T, Shijo R, Eguchi A, Tamaki H. Effects of aerodynamic interferences between heaving and torsional vibration of bridge decks: the case of Tacoma Narrows Bridge. *J Wind Eng Ind Aerodyn* 2003;91:1547–57.
- [38] Kumarasena T, Scanlan RH, Ehsan F. Wind induced motions of Deer Isle Bridge. *J Struct Eng* 1991;117:3356–74.
- [39] Battista RC, Pfeil MS. Reduction of vortex-induced oscillations of Rio-Niteroi bridge by dynamic control devices. *J Wind Eng Ind Aerodyn* 2000;84:273–88.
- [40] Corriols AS, Morgenthal G. Computational analysis of VIV observed on existing bridges. In: *The Seventh International Colloquium on Bluff Body Aerodynamics and Applications (BBAA7)*. Shanghai, China; 2012.
- [41] Schewe G, Larsen A. Reynolds number effects in the flow around a bluff bridge deck cross section. *J Wind Eng Ind Aerodyn* 1998;74–76:829–38.
- [42] Huang NE, Wu Z, Long SR, Arnold KC, Chen X, Blank K. On instantaneous frequency. *Adv Adaptive Data Anal* 2009;1:177–229.
- [43] Cao F, Ge Y. Air-induced nonlinear damping and added mass of vertically vibrating bridge deck section models under zero wind speed. *J Wind Eng Ind Aerodyn* 2017;169:217–31.
- [44] Gao G, Zhu L. Nonlinearity of mechanical damping and stiffness of a spring-suspended sectional model system for wind tunnel tests. *J Sound Vib* 2015;355:369–91.